ASSIGNMENT COVERSHEET

UTS: ENGINEERING & INFORMATION TECHNOLOGY

SUBJECT NUMBER & NAME	NAME OF STUDENT(s) (PRINT CLEARLY) SURNAME FIRST NAME	STUDENT ID(s)
41026 Advanced Software Development	*Lam Andy* *Hatzidimitriou Nicholas* *Lim Tristan* *Hein Htet Aung* *Barnard Matthew*	25249309 25471398 25299672 25026511 25563457

STUDENT EMAIL	STUDENT CONTACT NUMBER
andy.lam@student.uts.edu.au nicholas.b.hatzidimitriou@student.uts.edu.au tristan.j.lim@student.uts.edu.au heinhtet.aung@student.uts.edu.au matthew.k.barnard@student.uts.edu.au	0450615201 0477559936 0431765886 0421683687 0468962300

NAME OF TUTOR	TUTORIAL GROUP	DUE DATE
Georges Bou Ghantous	04	28/10/2026

ASSESSMENT ITEM NUMBER & TITLE

Assignment 3 - Release 2 Technical Report

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☑ I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet.
☑ I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements.
☑ I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension.

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Statement of collaboration:

Signature of student(s) Andy Lam Tristan Lim Nicholas Hatzidimitriou Hein Htet Aung Matthew Barnard
Date 28/10/2026

1. Project Overview

The Faculty Management System is a microservice application for a university faculty administration built by five students. Each student owns one stack feature (frontend, Flask backend and SQLite database), integrated behind one unified homepage and running together with one Docker Compose.

Student	Feature	Stack Folder
Andy Lam	Staff Management	student-1-Andy-Lam/
Hein Htet Aung	Teaching, Subject & Classroom Allocation	student-2Hein-Htet-Aung/
Tristan Lim	Workload & Availability Management	student-3Tristan-Lim/
Nicholas Hatzidimitriou	Research & Grant Management	student-4Nicholas-Hatzidimitriou/
Matthew Barnard	Performance & Professional Development	student-5Matthew-Barnard/

Release 1 delivered each student's individual CRUD stack with a shared MCP and RAG Server, AI-Mode and an Agentic review loop running on localhost.

Release 2 extends with:

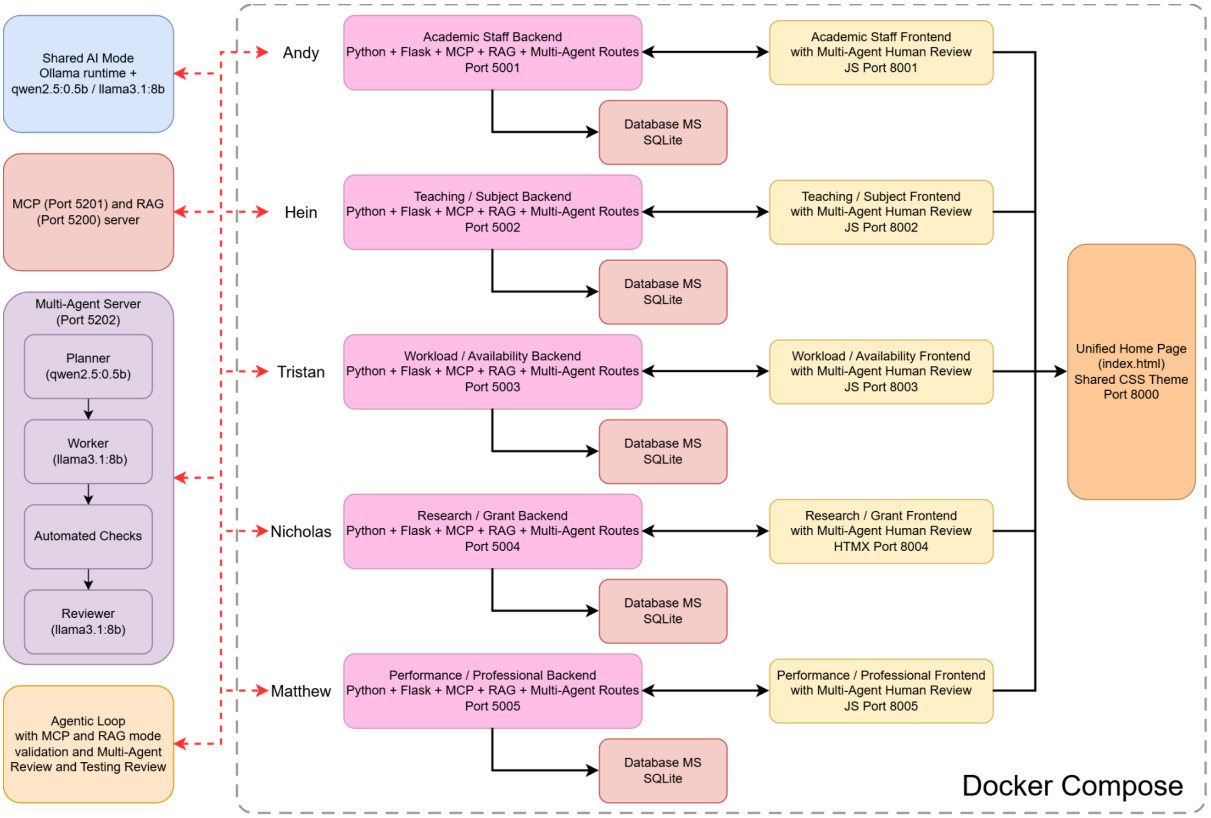
- A shared, host-run Multi-Agent Server (Planner, Worker, Reviewer, human review) with workflow history and audit logs, integrated into each individual student's backend and UI
- Three new agentic loop review modes: Multi-Agent Workflow, Testing Report and Cloud Deployment Review
- Individual student pre-commit security scans (Ruff, detect secrets and pip audit)
- Two post commit CI/CD endpoint tests per student, running through each student's individual workflow
- AWS cloud deployment configuration, with AI features disabled by default

2. Final Repository Structure



131 directories, 403 files

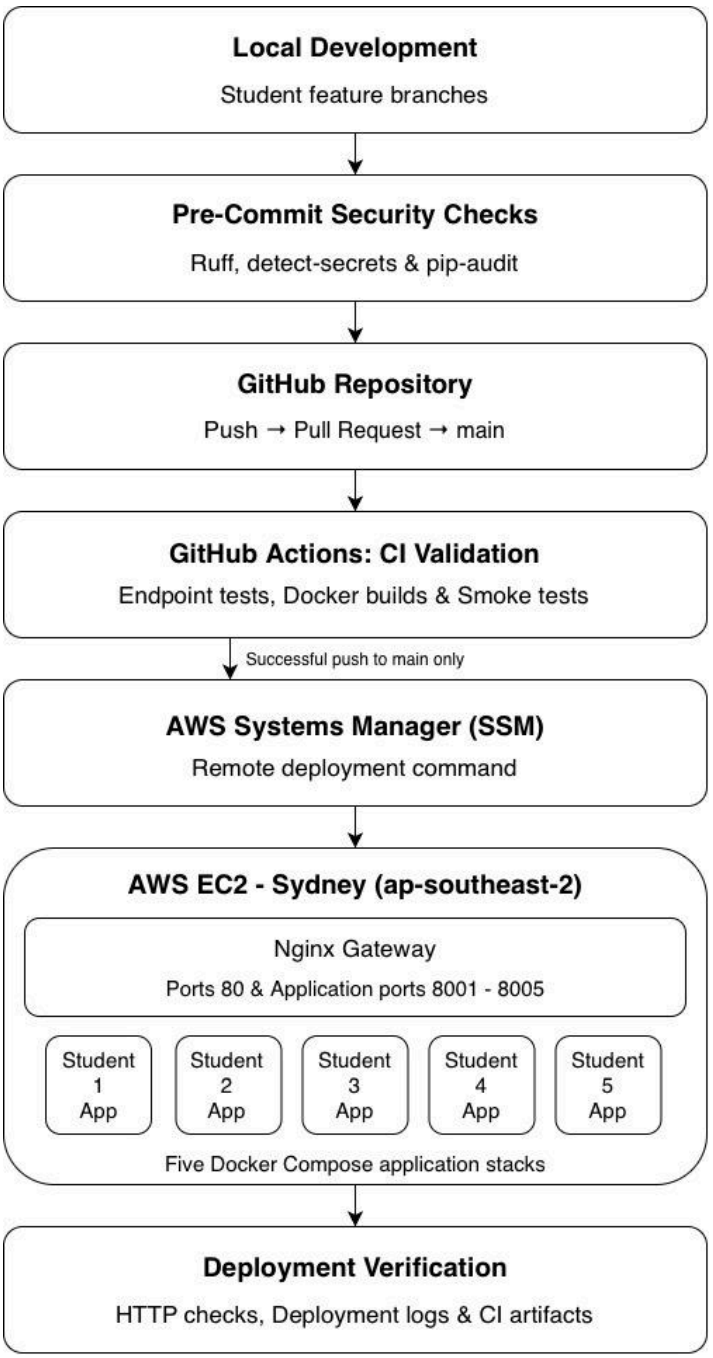
3. Release 2 Design Flow



Updated from release 1. Each feature's frontend calls its Flask backend, which reads and writes its own SQLite database service and the backend's AI-Mode, MCP and RAG routes call the shared Ollama runtime, MCP server (port 5201) and RAG server (port 5200). Release 2 adds a shared Multi-Agent server on port 5202 which runs outside Docker. A Multi-Agent request flows by, the user submits a request in the feature's frontend, the backend then forwards it to the Multi-Agent server and the Planner (qwen2.5:0.5b) sets the objective. The Worker (llama3.1:8b) answers using evidence from that feature's backend API, then automated checks confirm the output is complete and its citations exist and the Reviewer (llama3.1:8b) rates the risk and recommends a decision. The response returns through the backend to the frontend, where the user performs a Human Review by accepting, partially accepting, correcting or rejecting it and the decision would then be logged under the workflow ID.

4. Cloud Deployment and CI/CD Architecture

The Faculty Management System was deployed to AWS EC2 in the 'ap-southeast-2' region using Docker Compose and Nginx. This cloud configuration runs all five student applications, with Nginx serving as the entry point to the frontends and backend APIs.



GitHub Actions are used to validate the changes to be deployed in the CI/CD pipeline. Pre-commit security checks run locally and individual student workflows run automated endpoint tests, build Docker images, and run runtime smoke tests. The shared ‘cloud-deployment.yml’ workflow verifies the Docker Compose integration, Nginx routing, and service endpoints. Once validated, commits pushed to ‘main’ trigger deployment to EC2 via AWS Systems Manager (SSM). The ‘deploy/deploy.sh’ script then rebuilds and starts the containers and then checks the endpoints. GitHub Actions artifacts are stored for validation against deployment logs and results.

5. Pre-Commit Security Testing

Andy

- Three scans run on every commit: Ruff, detect-secrets and pip-audit.
- Ruff checks the backend, database and test code for security issues.
- Detect Secrets checks student 1 files for hardcoded keys, passwords and tokens.
- Pip-audit checks the backend dependencies in requirements.txt for vulnerabilities.
- Failure in scans causes the commit to be blocked until it is fixed.

```yaml
repos:
  - repo: local
    hooks:
      - id: student1-ruff-security
        name: Student 1 - Ruff Security Scan
        # S101 (assert) is how pytest checks results, so it is allowed in tests only
        entry: >-
          ruff check --select S
          --per-file-ignores 'student-1-Andy-Lam/tests/*.py:S101'
          --exclude student-1-Andy-Lam/tests/manual
          student-1-Andy-Lam/backend
          student-1-Andy-Lam/database
          student-1-Andy-Lam/tests
        language: python
        additional_dependencies: ["ruff==0.16.8"]
        pass_filenames: false
        always_run: true
        stages: [pre-commit]

      - id: student1-detect-secrets
        name: Student 1 - Secret Detection
        entry: detect-secrets-hook
        language: python
        additional_dependencies: ["detect-secrets==1.5.0"]
        files: ^student-1-Andy-Lam/
        types: [text]
        stages: [pre-commit]

      - id: student1-pip-audit
        name: Student 1 - Dependency Audit
        entry: >-
          pip-audit
          -r student-1-Andy-Lam/backend/requirements.txt
          --progress-spinner off
        language: python
        additional_dependencies: ["pip-audit==2.10.1"]
        pass_filenames: false
        always_run: true
        stages: [pre-commit]
```

```
(venv) C:\Users\kring\Documents\GitHub\faculty-management-system-ASD>pre-commit run -c student-1-Andy-Lam/.pre-commit-config.yaml --all-files --verbose
Student 1 - Ruff Security Scan.....................................Passed
- hook id: student1-ruff-security
- duration: 0.04s

All checks passed!

Student 1 - Secret Detection.....................................Passed
- hook id: student1-detect-secrets
- duration: 1.0s
Student 1 - Dependency Audit.....................................Passed
- hook id: student1-pip-audit
- duration: 26.67s

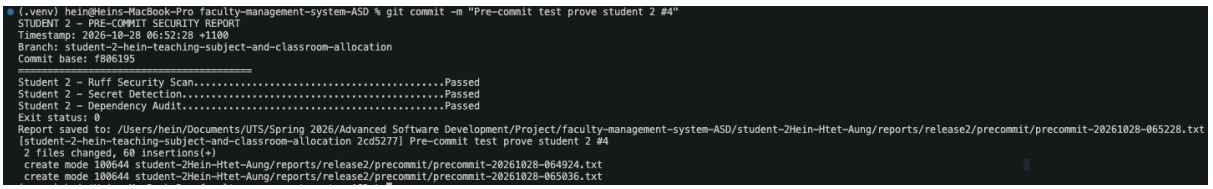
No known vulnerabilities found
```

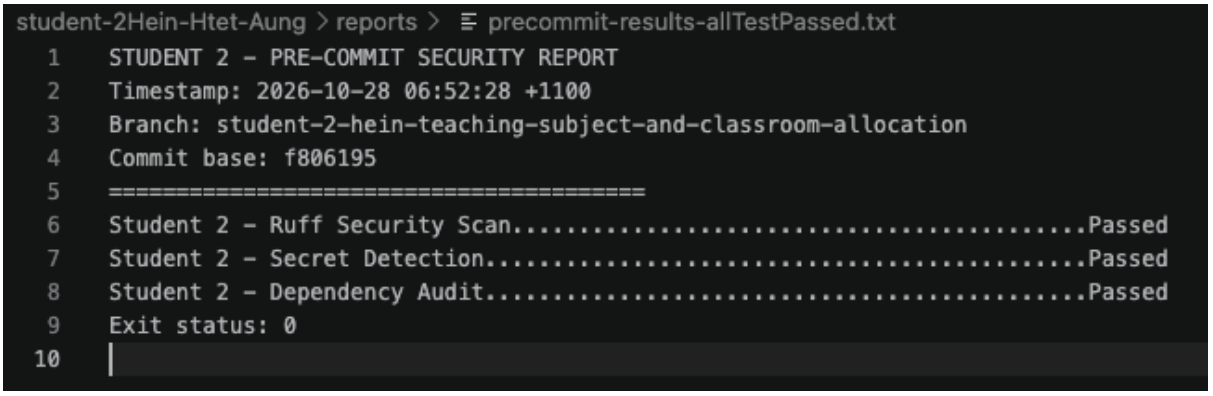
Hein

.pre-commit-config.yaml configuration is utilized to execute three security checks before each commit. Ruff scans the backend, database and test code for security vulnerabilities while detect-secrets checks staged text files for potential credentials. On the other hand, pip-audit checks the backend and database Python dependencies for known vulnerabilities.

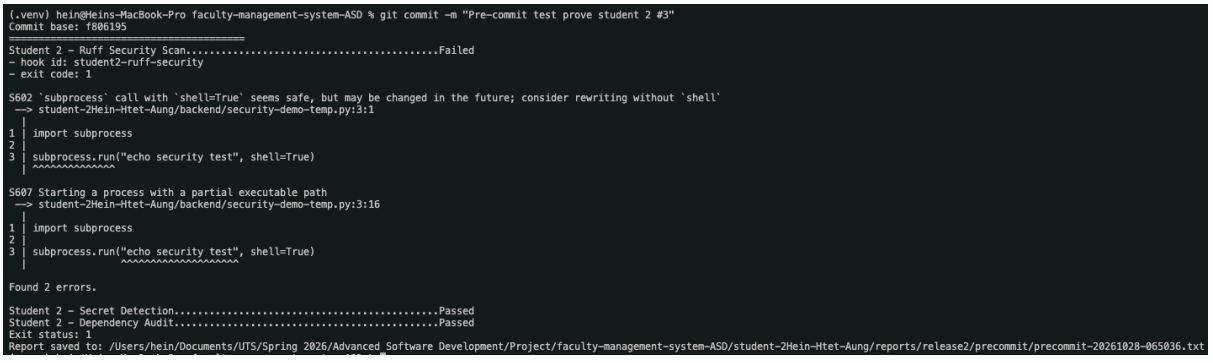
The configuration and execution results are retained in student-2Hein-Htet-Aung/, with the results stored under reports/release2/. The terminal output also provided evidence of the completed security checks.

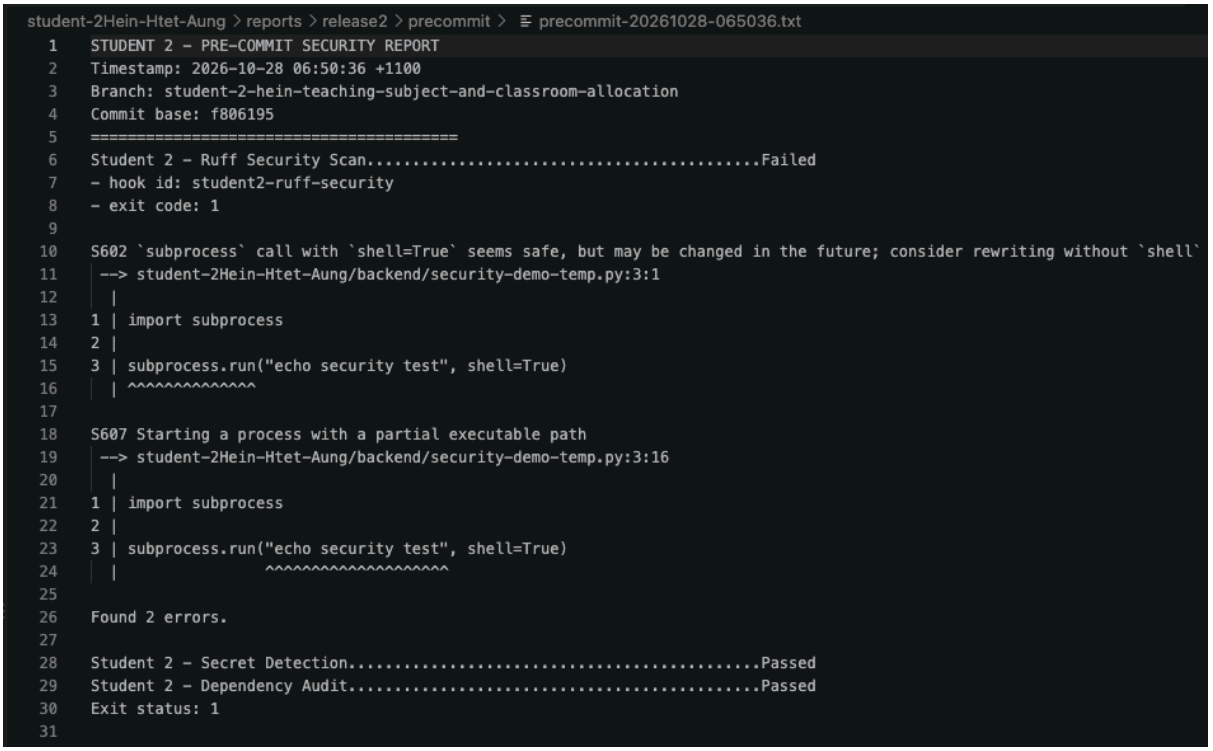
When all three security checks passed during the recorded execution:





When some of the security checks failed during the recorded execution (by including a code file that triggered the Ruff security scan):





.pre-commit-config.yaml configuration:



Tristan

The pre-commit hooks run Ruff security rules, detect secrets and pip-audit on my feature and on the shared Multi-Agent Server. The first scans found Flask debug mode on and vulnerable Flask. Flask-cors and requests versions, which I fixed by turning debug off by default and upgrading them. Ruff's SQL injection warnings were false positives as table and column names come from a schema allowlist and values are parameterised. All three hooks now pass on every commit.

```
PS D:\Coding Projects\faculty-management-system-ASD> .\.venv-run\Scripts\pre-commit.exe run -c student-3Tristan-Lim/.pre-commit-config.yaml --all-files
Student 3 - Ruff Security Scan.....................................Passed
Student 3 - Secret Detection.......................................Passed
Student 3 - Dependency Audit.......................................Passed
PS D:\Coding Projects\faculty-management-system-ASD>
```

Nicholas

The pre-commit hooks run Ruff security rules, detect secrets and pip audit on every commit. To show how a bad block of code would look, I created a file containing a fake secret and tried to commit it. The secret detection hook failed and the commit was rejected and the pre-commit test for secrets failed showing a detected secret. After removing the file, all three, Ruff, Secret Detection and Dependency Audit pass.

```
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$ git add student-4Nicholas-Hatzidimitriou/fake_secret_test.py
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$ git commit -m "planted fake secret to verify pre-commit blocking it"
[WARNING] Unstaged files detected.
[INFO] Stashing unstaged files to /home/nicho/.cache/pre-commit/patch1792896791-169898.
Student 4 - Ruff Security Scan.....................................Passed
Student 4 - Secret Detection.......................................Failed
- hook id: student4-detect-secrets
- exit code: 1

============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.1.1, pluggy-1.7.0 -- /home/nicho/.cache/pre-commit/repo3lj9m8gn/py_env-python3.14/bin/python
cachedir: .pytest_cache
rootdir: /home/nicho/faculty-management-system-ASD
collected 1 item

student-4Nicholas-Hatzidimitriou/tests/pre_commit/test_secrets_scan.py::test_detect_secrets_scan_passes FAILED [100%]

=================================== FAILURES ===================================
_____________________ test_detect_secrets_scan_passes _____________________

    def test_detect_secrets_scan_passes():
        result = subprocess.run(  # noqa: S603 - fixed detect-secrets invocation, not untrusted input
            [
                sys.executable, "-m", "detect_secrets", "scan",
                str(STUDENT4_ROOT.relative_to(REPO_ROOT)),
            ],
            capture_output=True,
            text=True,
            cwd=REPO_ROOT,
        )
        returncode = result.returncode
        assert returncode == 0, result.stdout + result.stderr

        current = json.loads(result.stdout)
        baseline = json.loads(BASELINE_PATH.read_text(encoding="utf-8"))

        audited_hashes = {
            (filename, entry["hashed_secret"])
            for filename, entries in baseline.get("results", {}).items()
            for entry in entries
            if entry.get("is_secret") is False
        }

        unaudited = {}
        for filename, entries in current.get("results", {}).items():
            new_entries = [
                entry for entry in entries
                if (filename, entry["hashed_secret"]) not in audited_hashes
            ]
            if new_entries:
                unaudited[filename] = new_entries

>       assert not unaudited, f"Unaudited potential secrets detected: {unaudited}"
E       AssertionError: Unaudited potential secrets detected: {'student-4Nicholas-Hatzidimitriou/fake_secret_test.py': [{'type': 'AWS Access Key', 'filename': 'student-4Nicholas-Hatzidimitriou/fake_secret_test.py', 'hashed_secret': '25910f981e85ca04baf359199dd0bd4a3ae738b6', 'is_verified': False, 'line_number': 2}, {'type': 'Secret Keyword', 'filename': 'student-4Nicholas-Hatzidimitriou/fake_secret_test.py', 'hashed_secret': '25910f981e85ca04baf359199dd0bd4a3ae738b6', 'is_verified': False, 'line_number': 2}]}
E       assert not {'student-4Nicholas-Hatzidimitriou/fake_secret_test.py': [{'type': 'AWS Access Key', 'filename': 'student-4Nicholas-Hatzidimitriou/fake_secret_test.py', 'hashed_secret': '25910f981e85ca04baf359199dd0bd4a3ae738b6', 'is_verified': False, ...}, {'type': 'Secret Keyword', 'filename': 'student-4Nicholas-Hatzidimitriou/fake_secret_test.py', 'hashed_secret': '25910f981e85ca04baf359199dd0bd4a3ae738b6', 'is_verified': False, ...}]}

student-4Nicholas-Hatzidimitriou/tests/pre_commit/test_secrets_scan.py:43: AssertionError
=========================== short test summary info ============================
FAILED student-4Nicholas-Hatzidimitriou/tests/pre_commit/test_secrets_scan.py::test_detect_secrets_scan_passes - AssertionError: Unaudited potential secrets detected: {'student-4Nicholas-H...
============================== 1 failed in 2.20s ===============================

Student 4 - Dependency Audit.....................................Passed
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$ git reset student-4Nicholas-Hatzidimitriou/fake_secret_test.py
Unstaged changes after reset:
M       student-4Nicholas-Hatzidimitriou/reports/release2/pre-commit-security-report.md
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$ rm student-4Nicholas-Hatzidimitriou/fake_secret_test.py
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$ pre-commit run -c student-4Nicholas-Hatzidimitriou/.pre-commit-config.yaml --all-files
Student 4 - Ruff Security Scan.....................................Passed
Student 4 - Secret Detection.......................................Passed
Student 4 - Dependency Audit.....................................Passed
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$
```

Student 4 Pre-Commit Security Testing Report

Date: 2026-10-25
Commit: d0cb6e57ea4af4fac530ed5168266dcc52e2145a
Student: Nicholas Hatzidimitriou (Student 4, Research & Grant Management)

What each scan checks (one line each)

- **Ruff Security Scan** (`--select S`) - static analysis for insecure code patterns:
  hardcoded binds, `debug=True`, SQL string-building, unsafe `subprocess`/`urlopen` calls.
- **Secret Detection** (`detect-secrets scan`) - entropy- and pattern-based scan for
  committed credentials (API keys, AWS keys, passwords), checked against an audited
  baseline so known false positives don't block every commit.
- **Dependency Audit** (`pip-audit`) - checks each `requirements.txt` against the
  PyPI Advisory Database for known CVEs in pinned package versions.

Current design: scoped to my own code only

The hook config lives at `student-4Nicholas-Hatzidimitriou/.pre-commit-config.yaml`
(moved there from the repo root to match students 1 and 2's layout). Because
`pre-commit install` only ever reads `.pre-commit-config.yaml` from the repo root, it
was reinstalled with:

```bash
pre-commit install -c student-4Nicholas-Hatzidimitriou/.pre-commit-config.yaml
```

— confirmed by reading the generated `.git/hooks/pre-commit` script, which now hardcodes
that `--config` path. Three hooks (`student4-ruff-security`, `student4-detect-secrets`,
`student4-pip-audit`), each scanning only:

- `student-4Nicholas-Hatzidimitriou/backend`
- `student-4Nicholas-Hatzidimitriou/database-service`
- `student-4Nicholas-Hatzidimitriou/frontend`
- `student-4Nicholas-Hatzidimitriou/tests`

Supporting files also moved into this folder: `tests/pre_commit/{test_ruff_scan,
test_secrets_scan,test_dependency_audit}.py`, `scripts/pre_commit_pytest.py`. The
shared root-level `pyproject.toml` was removed - ruff's `--select S` and
`--per-file-ignores` are now passed directly on the command line (same pattern as
students 1 and 2), so no shared ruff config is needed at all.

This replaced an earlier repo-wide design (one root `pyproject.toml` + `.pre-commit-config.yaml`
scanning all five stacks via `rglob`). That version correctly caught real findings (below),
but also meant `pre-commit run --all-files` failed on every commit attempt until all five
students fixed their own pre-existing issues - which defeats the ownership model in the
brief ("each student runs and reports on their own code"). Scoping it to my own folder
means my commits are gated by my own code's security posture only, not blocked by
findings I don't own and can't fix.

Fresh output (today, this scope)

```bash
pre-commit install -c student-4Nicholas-Hatzidimitriou/.pre-commit-config.yaml
pre-commit run -c student-4Nicholas-Hatzidimitriou/.pre-commit-config.yaml --all-files
```

```
Student 4 - Ruff Security Scan.....................................Passed
Student 4 - Secret Detection.......................................Passed
Student 4 - Dependency Audit.......................................Passed
```

Individually, for the record (same real run, split out):

```
$ ruff check --select S --per-file-ignores 'student-4Nicholas-Hatzidimitriou/**/tests/**:S101' \
    student-4Nicholas-Hatzidimitriou/{backend,database-service,frontend,tests}
All checks passed!

$ pip-audit -r student-4Nicholas-Hatzidimitriou/backend/requirements.txt
No known vulnerabilities found

$ pip-audit -r student-4Nicholas-Hatzidimitriou/database-service/requirements.txt
No known vulnerabilities found

$ pip-audit -r student-4Nicholas-Hatzidimitriou/frontend/requirements.txt
No known vulnerabilities found

$ detect-secrets scan student-4Nicholas-Hatzidimitriou
{"results": {}}
```

Zero findings, scoped scan. This isn't the absence of a check - it reflects real fixes
already applied (below) plus the fact a scoped scan no longer sees other students' or
teammates' shared-file findings, by design.

Real findings fixed (from the original repo-wide run, still true)

In **my own code**, all fixed:

File	Finding	Fix

```
|---|---|---|
| `student-4Nicholas-Hatzidimitriou/backend/app.py:36` | S104 (bind `0.0.0.0`) + S201 (`debug=True`) | `debug` now reads `FLASK_DEBUG` env var (default `false`); `noqa: S104` with reason - container needs to bind all interfaces |
| `student-4Nicholas-Hatzidimitriou/database-service/app.py:614` | Same | Same fix |
| `student-4Nicholas-Hatzidimitriou/frontend/app.py:37` | Same | Same fix |

**Before/after, `student-4Nicholas-Hatzidimitriou/backend/app.py`:**
```python
# Before
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=int(os.environ.get("PORT", 5004)), debug=True)

# After
if __name__ == "__main__":
    debug_mode = os.environ.get("FLASK_DEBUG", "false").lower() == "true"
    app.run(host="0.0.0.0", port=int(os.environ.get("PORT", 5004)), debug=debug_mode)  # noqa: S104 - container needs to bind all interfaces
```

In **shared `ai-services/` code I maintain** (not scanned by my scoped hook any more,
but still fixed in those files - this is independent of scoping, since the fixes are
in the files themselves):

File	Finding	Fix
`ai-services/mcp-server/server.py:12`	S104 (bind `0.0.0.0`)	`noqa: S104` - this host process must bind all interfaces so containerised backends can reach it via `host.docker.internal`
`ai-services/rag-server/rag_http_server.py:310`	Same	Same fix
`ai-services/rag-server/rag_pipeline.py:2357`	S110 (`except: pass`)	Replaced with `logger.warning(...)` - a real fix, not a suppression
`ai-services/agentic_loop/collectors/db_collector.py:35`	S608 (possible SQL injection)	`noqa: S608` - the table name comes from `sqlite_master`, not external input
`ai-services/agentic_loop/collectors/development_integrity_collector.py:47`	S608	`noqa: S608` - table/columns come from the module's own fixed `RESOURCES` config, not external input
`ai-services/agentic_loop/collectors/development_integrity_collector.py:90, 93`	S310 (`urlopen` scheme)	`noqa: S310` - fixed internal service URL, not external input

detect-secrets

Current scoped scan: zero results (see above). `.secrets.baseline` (repo root, shared)
still contains one older entry from before this scope change, in a teammate's Ollama
client file - the required placeholder value for an OpenAI-compatible client pointed at
a local Ollama server, not a credential - audited as `"is_secret": false`. My scoped
scan no longer reads that file at all, so this entry is inert for my own hook now, but
it's left in place since the baseline is a shared ledger other students' own scoped
hooks may still reference.

pip-audit

Fixed in my own three `requirements.txt` files (still true, re-verified today - see
fresh output above, all three now clean):

Package	Before	After	CVEs fixed
Flask	3.0.3	3.1.3	PYSEC-2026-2151
flask-cors	4.0.1	6.0.0	PYSEC-2024-71, PYSEC-2026-1383/1384/1385
requests	2.32.3	2.33.0	PYSEC-2026-1872, PYSEC-2026-2275
python-dotenv	1.0.1	1.2.2	PYSEC-2026-2270

This was a real upgrade, not just an edit - `flask-cors` 4→6 is a major version bump, so
all three Docker images were rebuilt and tested against the real database container
before trusting the change (`/health`, `/projects`, and a real CORS preflight request all
verified, then the containers removed).

Not fixed - no fix available upstream: `ai-services/rag-server/requirements.txt` pins
`chromadb` (unpinned, resolves to 1.5.9, the current latest release). `pip-audit` flags 4
CVEs against it (PYSEC-2026-311, -3813, -3814, -3815), none with a fix version yet.
Documented openly rather than suppressed; this is shared code, not mine to fix.

My scoped `pip-audit` hook no longer checks other students' `requirements.txt` files -
under the ownership model, that's each student's own responsibility, run through their
own scoped hook once they set one up (students 1 and 2 already have, independently).

Blocked-commit verification (re-run today, under the current scoped config)

Planted a fake AWS access key in a throwaway file inside my own scoped folder,
staged it, and attempted a real commit:

```bash
git add student-4Nicholas-Hatzidimitriou/fake_secret_test.py
git commit -m "test: deliberately planted fake secret to verify pre-commit blocks it"
```

Result: **blocked**.

```
Student 4 - Ruff Security Scan.....................................Passed
Student 4 - Secret Detection.......................................Failed
- hook id: student4-detect-secrets
- exit code: 1

FAILED student-4Nicholas-Hatzidimitriou/tests/pre_commit/test_secrets_scan.py::test_detect_secrets_scan_passes
AssertionError: Unaudited potential secrets detected: {'student-4Nicholas-Hatzidimitriou/fake_secret_test.py':
[{'type': 'AWS Access Key', ...}, {'type': 'Secret Keyword', ...}]}
```
```

```
Student 4 - Dependency Audit.....................................Passed
```

The scoped `student4-detect-secrets` hook caught it as both an `AWS Access Key` and a `Secret Keyword` finding, flagged it as unaudited, and the commit failed before it was created - `git log -1` still showed the prior commit (`d0cb6e5`), confirming nothing new was committed. The file was then unstaged and deleted, and the scan re-run clean:

```
Student 4 - Ruff Security Scan...................................Passed
Student 4 - Secret Detection.....................................Passed
Student 4 - Dependency Audit.....................................Passed
```

Summary

- Pre-commit scanning relocated from a repo-wide root config to one scoped entirely to `student-4Nicholas-Hatzidimitriou/`, matching students 1 and 2's layout, while remaining a real, auto-triggering git hook (verified via the installed hook script, not just manual runs).
- All three hooks pass cleanly today, scoped scan, fresh output captured above.
- 3 real findings fixed in my own code (S104 + S201, x3 files) - unchanged from before.
- 5 real/false-positive findings resolved in shared `ai-services/` code I maintain - unchanged from before, still true, just no longer covered by my own scoped hook.
- 4 real dependency vulnerabilities fixed across my own 3 `requirements.txt` files, re-verified clean today.
- 1 dependency vulnerability (chromadb) remains, documented as currently unfixable upstream, not mine to fix.
- Commit-blocking behaviour re-verified today with a real planted secret under the current scoped config, then reverted.

Matthew

My pre-commit hooks run Ruff security rules, detect-secrets and pip-audit. I tested secret detection by creating a temporary file containing a fake password and attempting to commit it. Secret Detection failed and blocked the commit. After removing the file, all three checks passed. I also updated vulnerable Flask and requests dependencies and documented the reviewed Ruff findings for container bindings and parameterised database queries.

```
(.venv) PS C:\Users\Matt\OneDrive\Documents\AdvSoftware\faculty-management-system-ASD> git commit -m "Student 5 security scan check"
[WARNING] Unstaged files detected.
[INFO] Stashing unstaged files to C:\Users\Matt\.cache\pre-commit\patch1793083905-15480.
Student 5 - Ruff Security Scan...................................Passed
Student 5 - Secret Detection.....................................Failed
- hook id: student5-detect-secrets
- exit code: 1

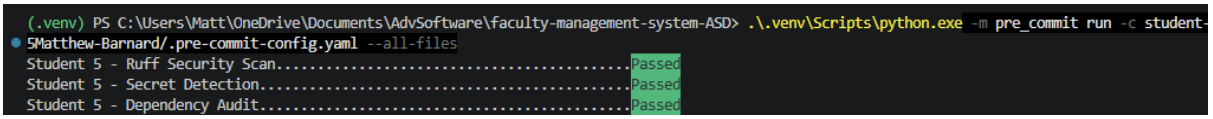
ERROR: Potential secrets about to be committed to git repo!

Secret Type: Secret Keyword
Location:    student-5Matthew-Barnard\security_scan_demo.py:1

Possible mitigations:
  - For information about putting your secrets in a safer place, please ask in
    #security
  - Mark false positives with an inline `pragma: allowlist secret` comment

If a secret has already been committed, visit
https://help.github.com/articles/removing-sensitive-data-from-a-repository

Student 5 - Dependency Audit.....................................Passed
[INFO] Restored changes from C:\Users\Matt\.cache\pre-commit\patch1793083905-15480.
(.venv) PS C:\Users\Matt\OneDrive\Documents\AdvSoftware\faculty-management-system-ASD>
```



The local commit hook runs Ruff security rules, detect-secrets and pip-audit. The scan covers Student 5 source and tests, including the Student 5 evidence contribution in the shared Multi-Agent server. Dependency auditing covers the backend and database requirements.

The first dependency audit found these advisories:

Package	Finding	Change
Flask 3.0.3	PYSEC-2026-2151	Updated both services to Flask 3.1.3
requests 2.32.3	PYSEC-2026-1872 and PYSEC-2026-2275	Updated the backend to requests 2.33.0

The initial Ruff output is saved in `ruff-initial.txt`. Its 12 findings were reviewed and the following exceptions are limited to the named files in `ruff.toml`:

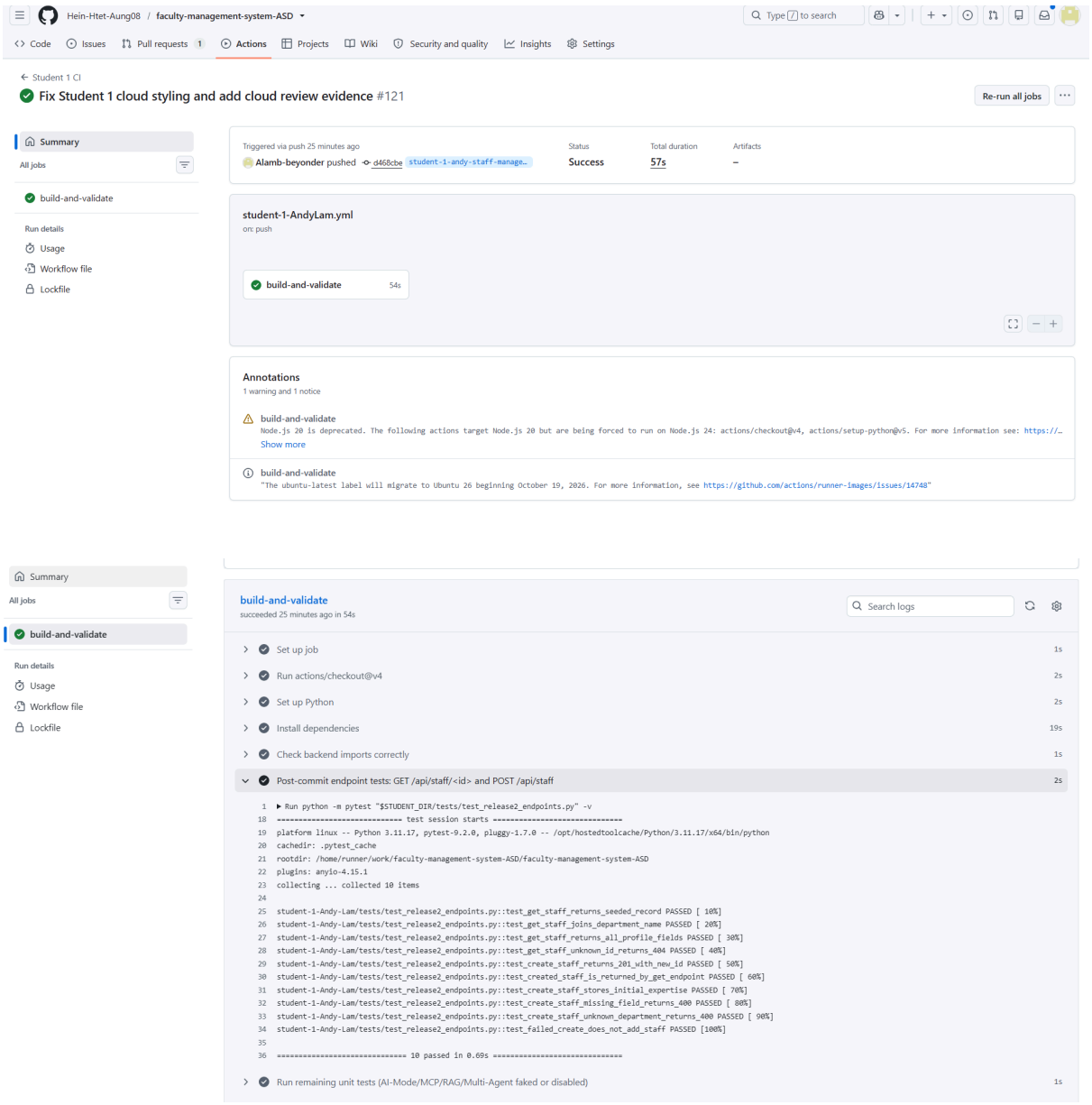
Rule	Location	Reason
S104	Backend and database `app.py`	Binding to 0.0.0.0 is needed for access from other containers. Debug remains off by default.
S608	Database `app.py`	Table, primary-key and column names come from the fixed resource configuration and validated field list. User values use SQL parameters.
S608	Database `database.py`	Seed and count queries use fixed table and column maps, with parameterised seed values.

S104 and S608 remain enabled elsewhere. The SQL-looking title test verifies that submitted text is stored as data and does not remove the goals table. These accepted findings are not claims that all SQL or network risks are absent.

6. Post-Commit CI/CD Testing

Andy

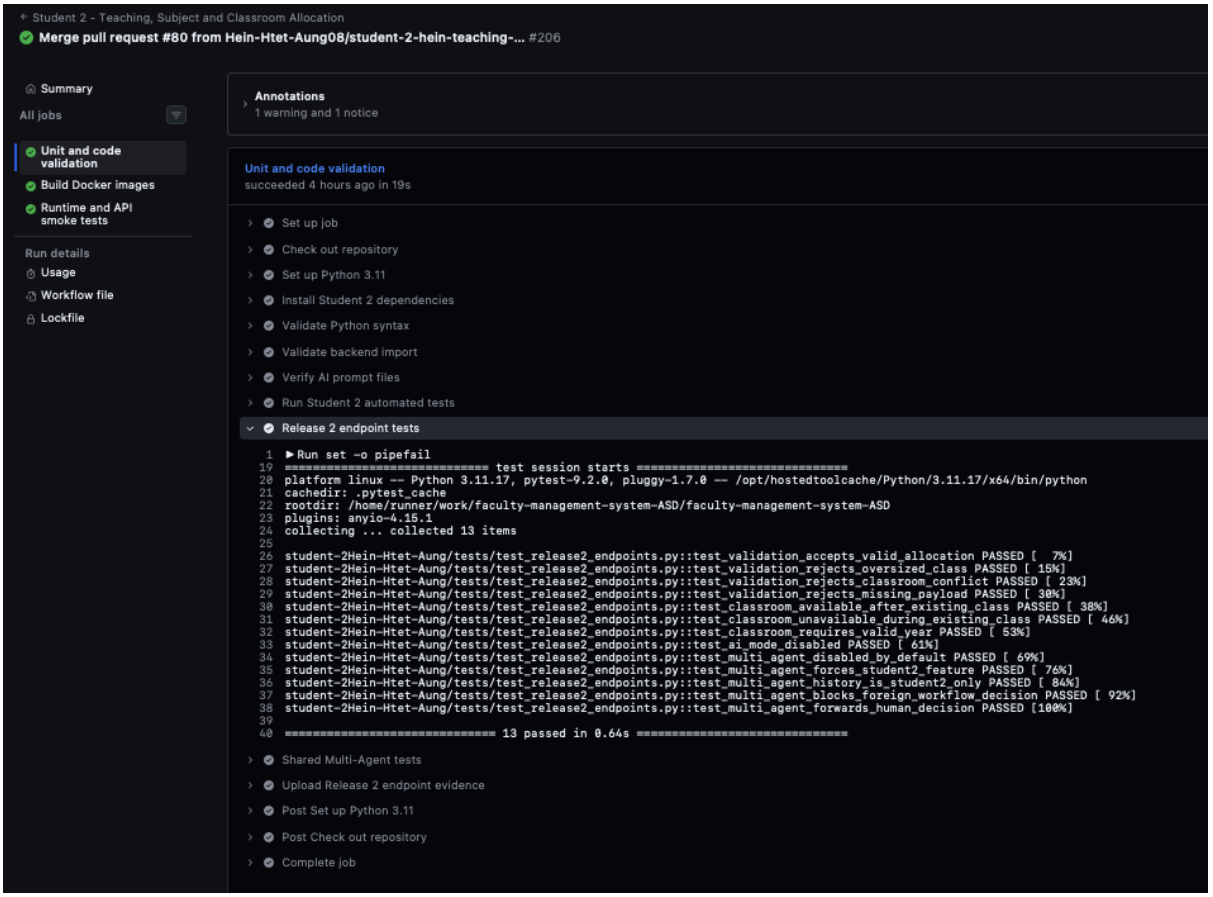
- Two backend endpoints are tested with pytest being GET /api/staff/<id> and POST /api/staff.
- In the GET tests, a seeded staff record is returned alongside its respective department name and profile fields. An unknown ID produces a 404.
- In the POST tests, a new staff member is created and a 201 is given.
- The created record can be read through the GET endpoint, missing fields or invalid department returns a 400 where failed requests don’t add a record to the database.
- The 64 unit tests run with AI-mode, MCP, RAG and Multi-Agent disabled.
- When the tests pass, Docker images are built for the frontend, backend and database services.
- Smoke tests check whether the services are reachable, CRUD functionality, and AI features return disabled responses in CI.
- Endpoint test results are saved in reports/release2/endpoint-tests.txt

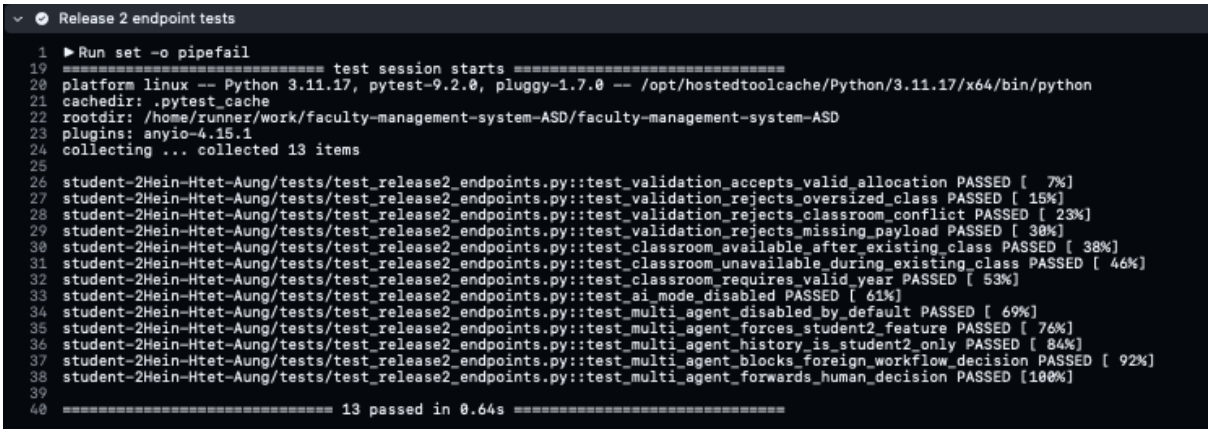


Hein

The GitHub Actions workflow “student-2-HeinHtetAung.yml” automates testing and validation after commits. Two backend/API endpoints, POST /teaching-allocations/validate and GET /classrooms/available, are tested using pytest. The tests include valid teaching allocations, classroom capacity restrictions, scheduling conflicts, missing request data and classroom availability.

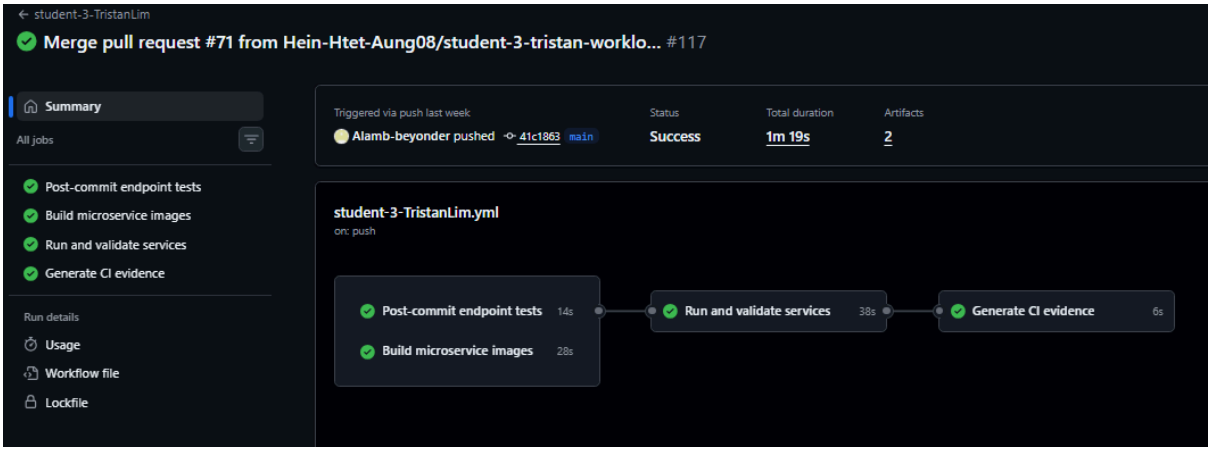
13 tests were executed successfully, including additional checks for AI-Mode and Multi-Agent integration. Once testing is successful, Docker images are built for the frontend, backend and database services. Then we do runtime smoke tests to check service accessibility and CRUD functionality. The results of the endpoint tests are automatically stored as a GitHub Actions artifact named `student-2-release2-endpoint-tests` after each workflow execution.





Tristan

After each push, my student-3-TristanLim.yml workflow tests my two JSON endpoints, Get /api/staff-workload/profiles and /alerts which the shared MCP, RAG and Multi-Agent servers read. Seven pytest tests check the returned records, filters, that alerts default to open and a 503 when the database is down. Four more run against the running containers and seeded data. The workflow also confirms MCP, RAG and Multi-Agent return 403 when disabled and publishes the results on the run summary and as an artifact.



Post-commit endpoint tests summary

Student 3 Post-Commit Endpoint Test Report

14 passed, 0 failed, 4 skipped (18 tests)

Endpoint function	Test	Result	Time
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_profiles_returns_all_seeded_staff`	SKIP	0.000s
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_profiles_filters_by_status`	SKIP	0.000s
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_alerts_default_to_open`	SKIP	0.000s
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_alerts_filter_resolved`	SKIP	0.000s
Multi-Agent routes (switch and Student 3 boundary)	`test_workflow_is_403_when_multi_agent_disabled`	PASS	0.777s
Multi-Agent routes (switch and Student 3 boundary)	`test_status_reports_disabled_without_calling_the_server`	PASS	0.014s
Multi-Agent routes (switch and Student 3 boundary)	`test_workflow_always_sends_the_student3_feature`	PASS	0.014s
Multi-Agent routes (switch and Student 3 boundary)	`test_blank_request_is_400`	PASS	0.013s
Multi-Agent routes (switch and Student 3 boundary)	`test_workflow_id_must_be_a_uuid`	PASS	0.013s
Multi-Agent routes (switch and Student 3 boundary)	`test_cannot_decide_another_features_workflow`	PASS	0.014s
Multi-Agent routes (switch and Student 3 boundary)	`test_decision_forwards_only_decision_fields`	PASS	0.015s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_profiles_returns_rows_as_json`	PASS	0.014s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_profiles_passes_status_and_department_filters`	PASS	0.013s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_profiles_returns_empty_list_when_nothing_matches`	PASS	0.014s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_profiles_returns_503_when_database_is_unreachable`	PASS	0.014s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_alerts_default_to_open_alerts`	PASS	0.015s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_alerts_accept_an_explicit_status`	PASS	0.014s
GET /api/staff-workload/profiles and /alerts (database faked)	`test_alerts_return_503_when_database_times_out`	PASS	0.014s

Run and validate services summary

Student 3 Post-Commit Endpoint Test Report

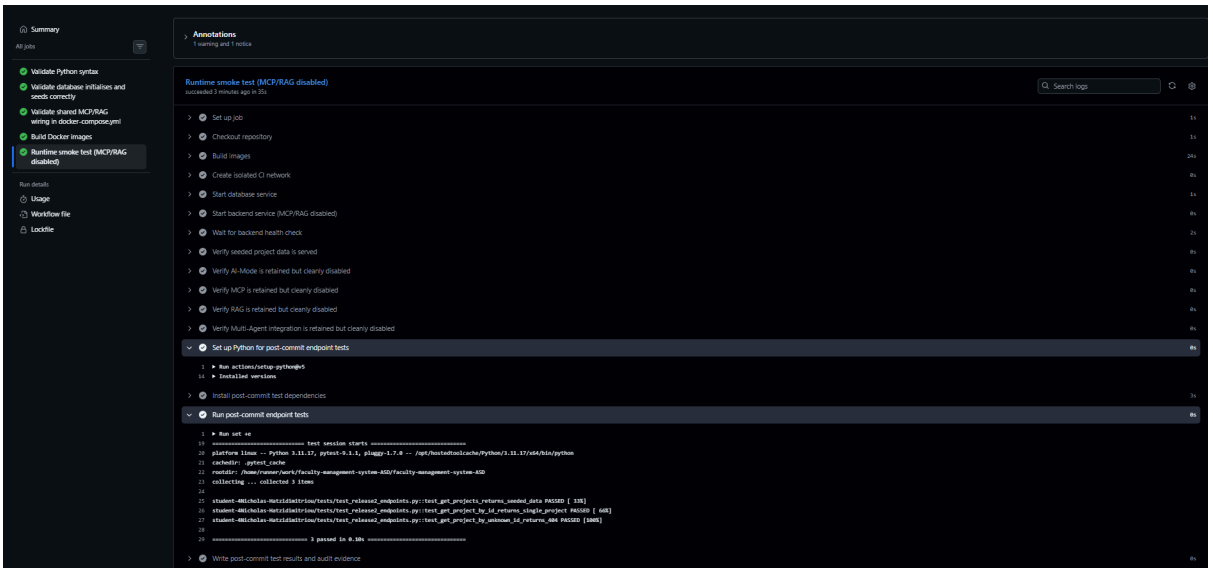
4 passed, 0 failed, 0 skipped (4 tests)

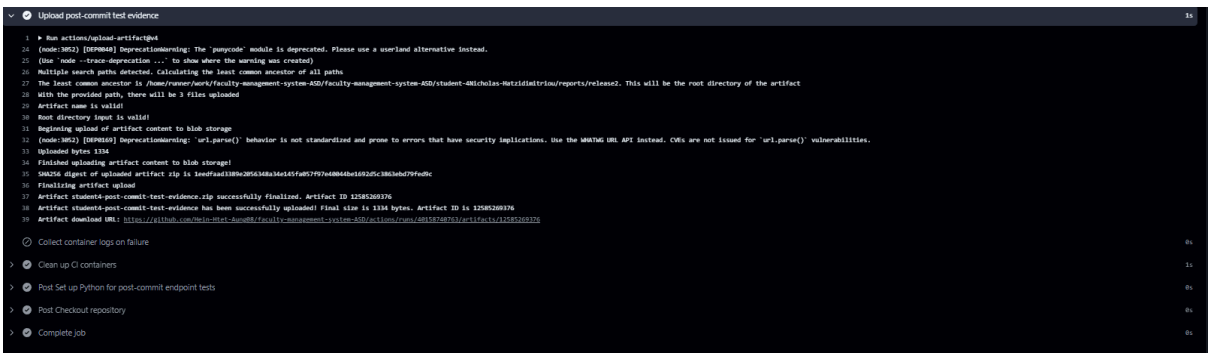
Endpoint function	Test	Result	Time
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_profiles_returns_all_seeded_staff`	PASS	0.014s
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_profiles_filters_by_status`	PASS	0.006s
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_alerts_default_to_open`	PASS	0.006s
GET /api/staff-workload/profiles and /alerts (running containers)	`test_live_alerts_filter_resolved`	PASS	0.005s

Job summary generated at run-time

Nicholas

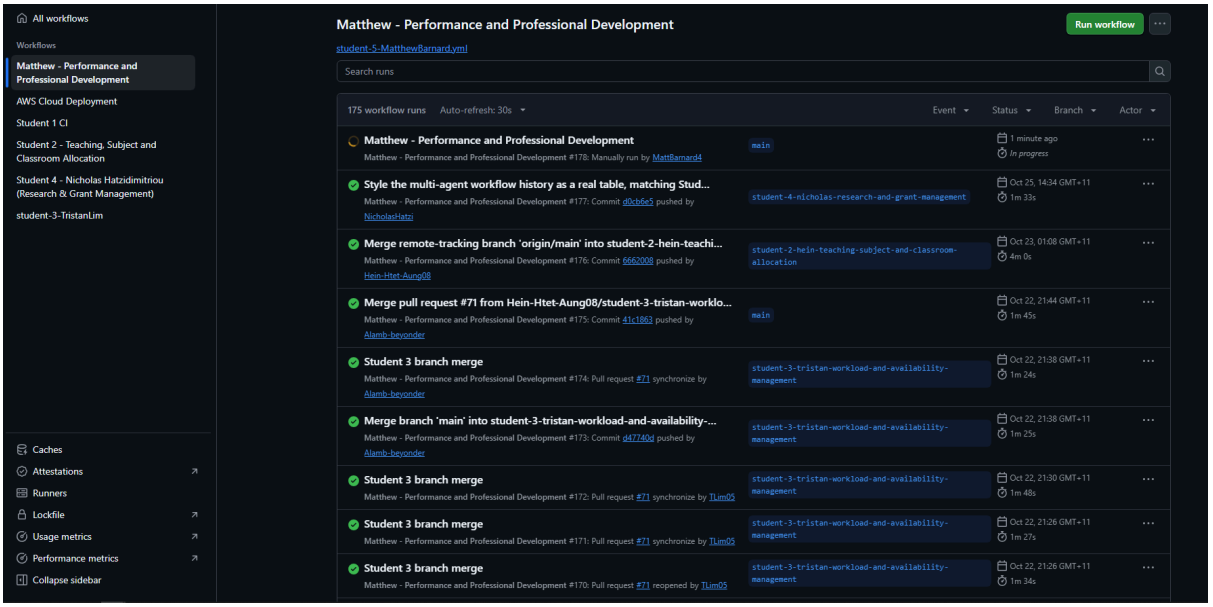
After each push, my student-4 workflow runs in Github Actions. The Runtime smoke test job starts the database and backend with MCP, RAG and Multi-Agent disabled, verifying that they are actually disabled and then runs the post-commit endpoint tests using pytest. The tests ensure ‘GET /projects’, ‘GET /projects/<id>’ and a request from an unknown ID that returns 404. The workflow then writes the audit record and uploads the results as an artifact.





Matthew

My GitHub Actions workflow tests GET and POST /api/development-goals using 14 tests. These cover returned records, filtering, persistence, invalid input, database failures and SQL-looking text being stored as data. Each test uses a temporary SQLite database and the real backend and database validation, with a test adapter replacing the HTTP connection. The workflow also builds the three containers and checks CRUD through the frontend proxy with all four AI modes disabled. A successful main-branch run is linked below, with endpoint results uploaded as text and JUnit XML.



7. Testing Review

The new addition to the shared Agentic Loop is option 10: Testing Report Review that collects the pre-commit and post-commit evidence for all five students. Each student has a CI configuration present, security results for Ruff, detect secrets and pip audit recorded as passing and a saved endpoint report. The verdict of the agentic loop being NOT_READY, was due to the saved evidence being unable to prove that the scans and tests were on the most current commit and Github actions were not queried in this. The model then reviewed and corrected the first assessment, which had claimed there were no unverified checks and recommended rerunning Github actions to confirm that the evidence files were there. The findings still remain authoritative and a human makes the release decision.

```
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD$ cd ai-services/agentic_loop/
nicho@DESKTOP-LJT9OTN:~/faculty-management-system-ASD/ai-services/agentic_loop$ python3 main.py
======================================================================
Prompt path map:
  db            (per-student) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/service
      - service/implementation/system_prompt.txt
      - service/implementation/task_prompt.txt
      - service/implementation/context_prompt.txt
  endpoints     (per-student) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/service
      - service/implementation/system_prompt.txt
      - service/implementation/task_prompt.txt
      - service/implementation/context_prompt.txt
  architecture  (whole-codebase) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/team
      - team/implementation/architecture_system_prompt.txt
      - team/implementation/architecture_task_prompt.txt
      - team/review/agent_review_prompt.txt
  ai_mode       (per-student) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/service
      - service/ai_mode/system_prompt.txt
      - service/ai_mode/task_prompt.txt
      - service/ai_mode/context_prompt.txt
  development_integrity (per-student) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/service
      - service/development_integrity/system_prompt.txt
      - service/development_integrity/task_prompt.txt
      - service/development_integrity/context_prompt.txt
  mcp           (whole-codebase) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/lab7
      - lab7/implementation/mcp_implementation_prompt.txt
      - lab7/review/mcp_review_prompt.txt
      - lab7/review/mcp_reasoning_prompt.txt
  rag           (whole-codebase) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/lab8
      - lab8/implementation/rag_implementation_prompt.txt
      - lab8/review/rag_review_prompt.txt
      - lab8/review/rag_reasoning_prompt.txt
  multi_agent_workflow_review (whole-codebase) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/release2
      - release2/implementation/multi_agent_workflow_prompt.txt
      - release2/review/multi_agent_workflow_review_prompt.txt
  testing_report_review (whole-codebase) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/release2
      - release2/implementation/testing_report_prompt.txt
      - release2/review/testing_report_review_prompt.txt
  cloud_deployment_report_review (whole-codebase) -> /home/nicho/faculty-management-system-ASD/ai-services/agentic_loop/prompts/release2
      - release2/implementation/cloud_deployment_report_prompt.txt
      - release2/review/cloud_deployment_report_review_prompt.txt

======================================================================
Agentic Loop
  1) DB             - validate every student's database (real queries)
  2) Endpoints      - test every student's live endpoints (real HTTP)
  3) Architecture   - file layout + docker-compose (two-model review)
  4) AI-Mode        - review every student's Ollama integration (static analysis)
  5) Development Integrity (Student 5) - review development goal integrity
  6) MCP            - validate shared MCP integration
  7) RAG            - validate shared RAG integration
  8) Run All
  9) Multi-Agent Workflow Review
 10) Testing Report Review
 11) Cloud Deployment Report Review
  0) Exit
======================================================================
Select an option> 10

Testing Report Review:
  1) Review saved results (quick; no tests or network)
  2) Run fresh local tests + check GitHub Actions (may take minutes)
Choose 1 or 2 [1]> 1

Running Testing Report Review

[TESTING REPORT REVIEW][OBSERVE] Collecting testing evidence.
[TESTING REPORT REVIEW][OBSERVE] Testing evidence collected.
[TESTING REPORT REVIEW][ACT] Implementation assessment complete.
[TESTING REPORT REVIEW][ADAPT] Independent review complete.

OBSERVE:
TESTING REPORT REVIEW EVIDENCE
Collection mode: SAVED
Students inspected: 5
Students requiring more evidence: [1, 2, 3, 4, 5]
DETERMINISTIC VERDICT: NOT_READY
This verdict uses test exit codes and available artifacts, not AI judgement.
CI success requires a run at the EXACT current Git commit.
Saved security scan results do not prove scans ran at the current commit.
CI verification: GitHub Actions was not queried in saved-report mode.
Saved evidence files found: 10
Saved test results are NOT proven current merely by matching test counts.
No fresh endpoint tests were executed.

Student 1: REVIEW_REQUIRED
  directory: student-1-Andy-Lam
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 10 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 2: REVIEW_REQUIRED
  directory: student-2Hein-Htet-Aung
```

```
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 13 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 3: REVIEW_REQUIRED
  directory: student-3Tristan-Lim
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 7 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 4: REVIEW_REQUIRED
  directory: student-4Nicholas-Hatzidimitriou
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 3 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 5: REVIEW_REQUIRED
  directory: student-5Matthew-Barnard
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 14 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED


IMPLEMENTATION ASSESSMENT:
Observed Evidence:
- All students have a present CI configuration but no CI execution was checked in saved mode.
- Security checks (ruff, detect-secrets, pip-audit) are recorded as PASS_RECORDED for all students.
- Local tests passed for each student with varying numbers of tests: 10, 13, 7, 3, and 14 respectively.
- Saved endpoint reports are recorded as PASS_RECORDED for all students.

Unverified or Failed Checks:
```

```
Unverified or Failed Checks:
- No specific unverified or failed checks were reported in the evidence provided.

Security Findings:
- All security checks (ruff, detect-secrets, pip-audit) passed for each student. No security findings were identified.

Recommended Human Follow-up:
- Verify that the exact current Git commit was used for the saved reports and tests.
- Ensure that GitHub Actions CI execution is checked in a fresh run to confirm its success at the current commit.
- Confirm that all saved evidence files are up-to-date with the current commit.

INDEPENDENT REVIEW:
**Supported Statements:**

* All students have a present CI configuration.
* Security checks (ruff, detect-secrets, pip-audit) are recorded as PASS_RECORDED for all students.
* Local tests passed for each student with varying numbers of tests: 10, 13, 7, 3, and 14 respectively.
* Saved endpoint reports are recorded as PASS_RECORDED for all students.

**Unsupported or Contradicted Claims:**

* The FIRST MODEL ASSESSMENT claims that no specific unverified or failed checks were reported in the evidence provided. However, the ORIGINAL TESTING EVIDENCE states that CI execution was not checked
in saved mode (ci_execution: NOT_CHECKED_IN_SAVED_MODE) for all students.
* The FIRST MODEL ASSESSMENT implies that security checks passed for each student without verifying the exact current Git commit used for the saved reports and tests. However, the ORIGINAL TESTING EVID
ENCE warns that saved security scan results do not prove scans ran at the current commit.

**Remaining Gaps:**

* Verify that the exact current Git commit was used for the saved reports and tests.
* Ensure that GitHub Actions CI execution is checked in a fresh run to confirm its success at the current commit.
* Confirm that all saved evidence files are up-to-date with the current commit.

**Recommended Human Follow-up:**

* Investigate why CI execution was not checked in saved mode (ci_execution: NOT_CHECKED_IN_SAVED_MODE) for all students.
* Verify the security checks' results by re-running them at the exact current Git commit to confirm their PASS_RECORDED status.
[TESTING REPORT REVIEW][RECORD] Report saved: /home/nicho/faculty-management-system-ASD/reports/release2/testing_report_review-20261025-151243.md
Testing Report Review completed.
Report: /home/nicho/faculty-management-system-ASD/reports/release2/testing_report_review-20261025-151243.md
Deterministic findings remain authoritative.

======================================================================
```

Validation Log

reports > release2 > {} testing_report_review-20261025-150759.json > ...


```
{
  "mode": "testing_report_review",
  "label": "Testing Report Review",
  "timestamp": "20261025-150759",
  "evidence": "TESTING REPORT REVIEW EVIDENCE\nCollection mode: SAVED\nStudents inspected: 5\nStudents requiring more evidence: [1, 2, 3, 4, 5]\nDETERMINISTIC VERDICT: NOT_READY
  "implementation_assessment": "### Observed Evidence:\n- **Student 1:** \n  - **CI Configuration:** PRESENT\n  - **CI Execution:** NOT_CHECKED_IN_SAVED_MODE\n  - **Security Sta
  "independent_review": "**Supported Statements:**\n\n* The ORIGINAL TESTING EVIDENCE is authoritative and should be used to verify all claims.\n* Student 1, 2, 3, 4, and 5 requ
  "human_validation_required": true
}
```

Review Log

reports > release2 > testing_report_review-20261025-150759.md > # Testing Report Review

```
# Testing Report Review

Generated: 20261025-150759

## OBSERVE: Collected Evidence

```text
TESTING REPORT REVIEW EVIDENCE
Collection mode: SAVED
Students inspected: 5
Students requiring more evidence: [1, 2, 3, 4, 5]
DETERMINISTIC VERDICT: NOT_READY
This verdict uses test exit codes and available artifacts, not AI judgement.
CI success requires a run at the EXACT current Git commit.
Saved security scan results do not prove scans ran at the current commit.
CI verification: GitHub Actions was not queried in saved-report mode.
Saved evidence files found: 10
Saved test results are NOT proven current merely by matching test counts.
No fresh endpoint tests were executed.

Student 1: REVIEW_REQUIRED
  directory: student-1-Andy-Lam
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 10 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 2: REVIEW_REQUIRED
  directory: student-2Hein-Htet-Aung
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 13 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 3: REVIEW_REQUIRED
  directory: student-3Tristan-Lim
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 7 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 4: REVIEW_REQUIRED
  directory: student-4Nicholas-Hatzidimitriou
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 3 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

Student 5: REVIEW_REQUIRED
  directory: student-5Matthew-Barnard
  ci_configuration: PRESENT
  ci_execution: NOT_CHECKED_IN_SAVED_MODE
  security_status: PASS_RECORDED
  security_details: {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  local_tests: PASS_RECORDED
  suite_details: 14 tests passed in saved snapshot; commit unverified
  saved_endpoint_report: PASS_RECORDED

```

ACT: Implementation Assessment

Observed Evidence:
- **Student 1:**
  - **CI Configuration:** PRESENT
  - **CI Execution:** NOT_CHECKED_IN_SAVED_MODE
  - **Security Status:** PASS_RECORDED
  - **Security Details:** {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  - **Local Tests:** PASS_RECORDED
  - **Suite Details:** 10 tests passed in saved snapshot; commit unverified

- **Student 2:**
  - **CI Configuration:** PRESENT
  - **CI Execution:** NOT_CHECKED_IN_SAVED_MODE
  - **Security Status:** PASS_RECORDED
  - **Security Details:** {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  - **Local Tests:** PASS_RECORDED
  - **Suite Details:** 13 tests passed in saved snapshot; commit unverified

- **Student 3:**
  - **CI Configuration:** PRESENT
  - **CI Execution:** NOT_CHECKED_IN_SAVED_MODE
  - **Security Status:** PASS_RECORDED
  - **Security Details:** {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', 'pip-audit': 'PASS_RECORDED'}
  - **Local Tests:** PASS_RECORDED
  - **Suite Details:** 7 tests passed in saved snapshot; commit unverified

- **Student 4:**
  - **CI Configuration:** PRESENT
  - **CI Execution:** NOT_CHECKED_IN_SAVED_MODE
  - **Security Status:** PASS_RECORDED
  - **Security Details:** {'ruff': 'PASS_RECORDED', 'detect-secrets': 'PASS_RECORDED', '
```

```
## ADAPT: Independent Review

**Supported Statements:**

* The ORIGINAL TESTING EVIDENCE is authoritative and should be used to verify all claims.
* Student 1, 2, 3, 4, and 5 require more evidence (REVIEW_REQUIRED).
* The deterministic verdict is NOT_READY.

**Unsupported or Contradicted Claims:**

* The FIRST MODEL ASSESSMENT's test counts for each student are not verified against the ORIGINAL TESTING EVIDENCE. Specifically:
  + Student 1: 10 tests passed, but no fresh endpoint tests were executed.
  + Student 2: 13 tests passed, but no fresh endpoint tests were executed.
  + Student 3: 7 tests passed, but no fresh endpoint tests were executed.
  + Student 4: 3 tests passed, but no fresh endpoint tests were executed.
  + Student 5: 14 tests passed, but no fresh endpoint tests were executed.
* The FIRST MODEL ASSESSMENT's claims of CI success are unsupported because the ORIGINAL TESTING EVIDENCE indicates that GitHub Actions was not queried in saved-report mode.

**Remaining Gaps:**

* Fresh endpoint tests were not executed for any student.
* Security scan results do not prove scans ran at the current commit.
* The security status and details for each student should be verified against the ORIGINAL TESTING EVIDENCE.

**Recommended Human Follow-up:**

* Verify the test counts and suite details for each student against the ORIGINAL TESTING EVIDENCE.
* Investigate why fresh endpoint tests were not executed for any student.
* Review the security scan results to ensure they are up-to-date and accurate.
* Suggest evidence collection and focused fixes, rather than model retraining.

## Human Validation

AI findings are advisory. The recorded workflow decisions remain authoritative. No source workflow or audit records were modified by this review.
```

8. Local Deployment Evidence

The application uses the root Docker Compose configuration to connect each feature's frontend, backend and database. Student 5 is available locally on port 8005 and sends API requests through Nginx to its backend on port 5005 and database service on port 5105. The Multi-Agent server runs separately on the host at port 5202 and reads feature evidence through backend APIs. In the recorded Student 5 execution, the Planner, Worker and Reviewer completed successfully before the human correction was saved. Local container checks also verified frontend assets, resource reads and a create-update-delete lifecycle. Workflow history and coordination logs identify the execution and decision.

All containers running:

```
(.venv) hein@Heins-MacBook-Pro faculty-management-system-ASD % docker compose ps
NAME                                        IMAGE                                                COMMAND                  SERVICE                CREATED         STATUS                   PORTS
faculty-management-system-asd-allocation-backend-1   faculty-management-system-asd-allocation-backend     "python app.py"          allocation-backend     12 hours ago    Up 12 hours              0.0.0.0:5002->5002/tcp, [::]:5002->5002/
tcp
faculty-management-system-asd-allocation-database-1  faculty-management-system-asd-allocation-database    "python app.py"          allocation-database    12 hours ago    Up 12 hours              6002/tcp
faculty-management-system-asd-allocation-frontend-1  sha256:8c0ddee6565a5653457f653cdd383a848eeeca59d9ce71860a65c4cec60c8a66   "/docker-entrypoint.…"   allocation-frontend    5 days ago      Up 12 hours              0.0.0.0:8002->80/tcp, [::]:8002->80/tcp
faculty-management-system-asd-staff-backend-1        faculty-management-system-asd-staff-backend          "python app.py"          staff-backend          12 hours ago    Up 12 hours              0.0.0.0:5001->5001/tcp, [::]:5001->5001/
tcp
faculty-management-system-asd-staff-database-1       faculty-management-system-asd-staff-database         "tail -f /dev/null"      staff-database         12 hours ago    Up 12 hours
faculty-management-system-asd-staff-frontend-1       sha256:346e57de7a9052bd54ef3f4b4bf6563f93ccd38836291e974690831a38cfab21   "/docker-entrypoint.…"   staff-frontend         6 days ago      Up 12 hours              0.0.0.0:8001->80/tcp, [::]:8001->80/tcp
ollama                                      ollama/ollama:latest                                 "/bin/ollama serve"      ollama                 3 weeks ago     Up 12 hours              0.0.0.0:11434->11434/tcp, [::]:11434->11
434/tcp
student3-backend                            faculty-management-system-asd-student3-backend       "python backend-serv…"   student3-backend       12 hours ago    Up 12 hours              0.0.0.0:5003->5003/tcp, [::]:5003->5003/
tcp
student3-database                           faculty-management-system-asd-student3-database      "python app.py"          student3-database      12 hours ago    Up 12 hours              0.0.0.0:5103->5103/tcp, [::]:5103->5103/
tcp
student3-frontend                           faculty-management-system-asd-student3-frontend      "/docker-entrypoint.…"   student3-frontend      12 hours ago    Up 12 hours              0.0.0.0:8003->80/tcp, [::]:8003->80/tcp
student4-backend                            faculty-management-system-asd-student4-backend       "python app.py"          student4-backend       12 hours ago    Up 12 hours              0.0.0.0:5004->5004/tcp, [::]:5004->5004/
tcp
student4-database                           faculty-management-system-asd-student4-database      "python app.py"          student4-database      12 hours ago    Up 12 hours              0.0.0.0:5104->5104/tcp, [::]:5104->5104/
tcp
student4-frontend                           faculty-management-system-asd-student4-frontend      "./docker-entrypoint.…"  student4-frontend      12 hours ago    Up 12 hours              0.0.0.0:8004->8004/tcp, [::]:8004->8004/
tcp
student5-backend                            faculty-management-system-asd-student5-backend       "python app.py"          student5-backend       12 hours ago    Up 12 hours (healthy)    0.0.0.0:5005->5005/tcp, [::]:5005->5005/
tcp
student5-database                           faculty-management-system-asd-student5-database      "python app.py"          student5-database      12 hours ago    Up 12 hours (healthy)    0.0.0.0:5105->5105/tcp, [::]:5105->5105/
tcp
student5-frontend                           sha256:0eb69989a2249665194ec4f5bf9d8f922dc1b9f7c35b996812e0ed26abb440ca   "/docker-entrypoint.…"   student5-frontend      6 days ago      Up 12 hours              0.0.0.0:8005->80/tcp, [::]:8005->80/tcp
unified-home                                nginx:alpine                                         "/docker-entrypoint.…"   home                   3 weeks ago     Up 12 hours              0.0.0.0:8000->80/tcp, [::]:8000->80/tcp
```

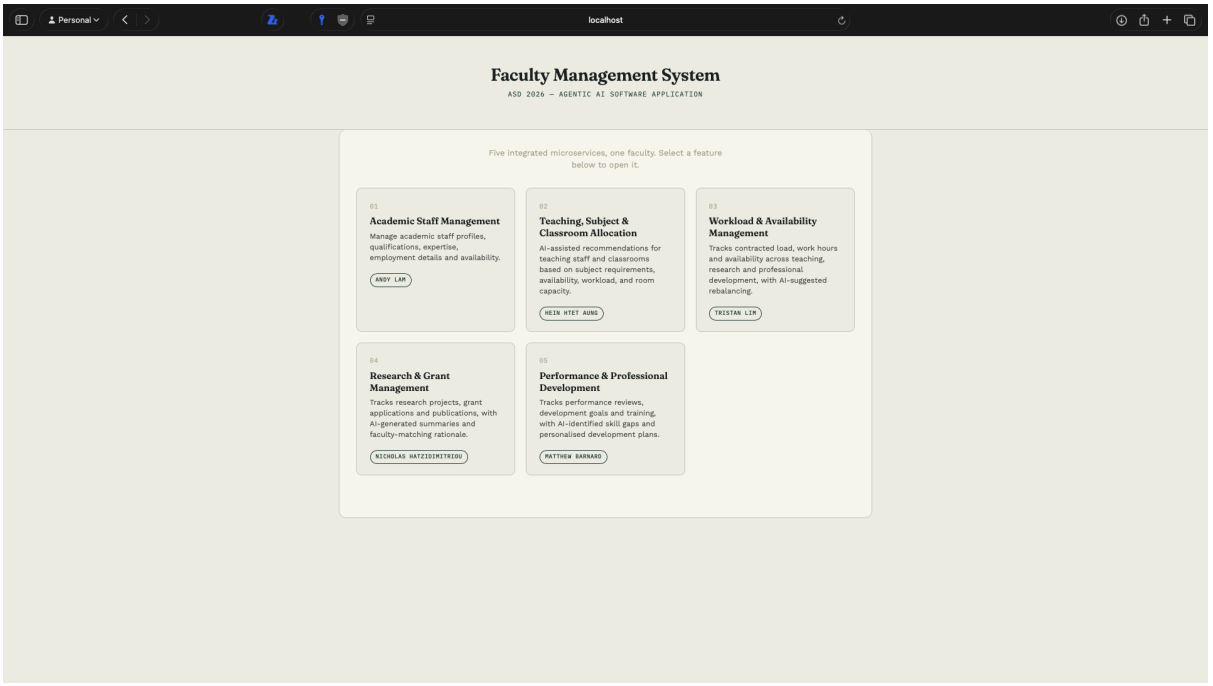
Shared AI/Multi-Agent services running:

```
(.venv) hein@Heins-MacBook-Pro mcp-server % python3 server.py
Starting Faculty Management System MCP Server...
Server status: RUNNING
Available tools:
- student1_get_staff_profile
- student1_search_staff_by_expertise
- student2_validate_teaching_allocation
- student2_check_classroom_availability
- student3_staff_count
- student3_staff_by_status
- student3_staff_workload_detail
- student3_open_alerts
- student4_project_count
- student4_projects_by_department
- student4_project_grants_summary
- student4_research_history
- student5_staff_development_summary
- student5_training_by_skill_area
INFO:     Started server process [92766]
INFO:     Waiting for application startup.
[10/28/26 11:14:34] INFO     StreamableHTTP session manager started
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:5201 (Press CTRL+C to quit)
```

```
(.venv) hein@Heins-MacBook-Pro rag-server % python3 rag_http_server.py
RAG HTTP server running on 0.0.0.0:5200
127.0.0.1 - - [28/Oct/2026 22:59:40] "POST /refresh HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 22:59:43] "POST /retrieve HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 23:01:13] "POST /answer HTTP/1.1" 200 -
```

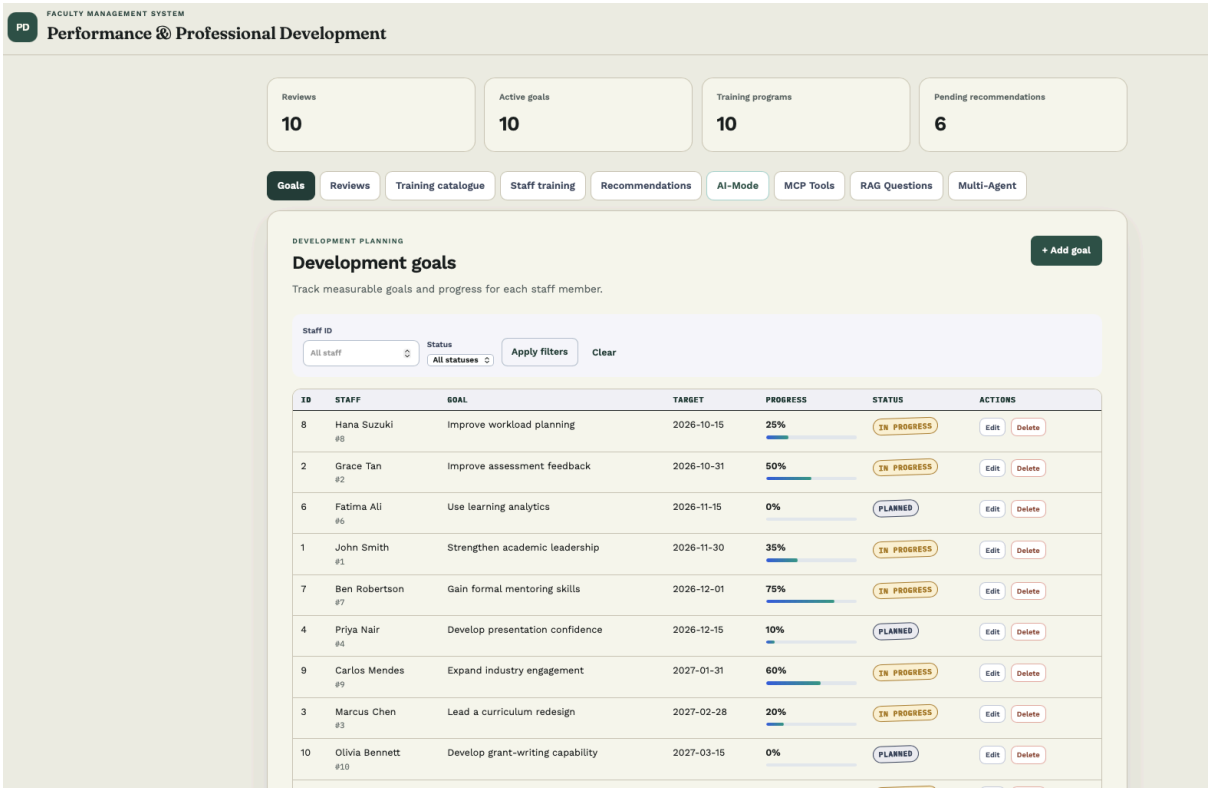
```
(.venv) hein@Heins-MacBook-Pro multi-agent-server % python3 app.py
multi-agent-server listening on http://0.0.0.0:5202
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5202
 * Running on http://192.168.1.152:5202
Press CTRL+C to quit
127.0.0.1 - - [28/Oct/2026 11:17:25] "GET /workflow/history?feature=student2&limit=20 HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 11:17:26] "GET /workflow/history?feature=student2&limit=20 HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 11:44:23] "POST /workflow HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 11:44:23] "GET /workflow/history?feature=student2&limit=20 HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 11:47:14] "GET /workflow/a91e5917-d7a9-4c4c-808e-992764117d2a HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 11:47:14] "POST /workflow/a91e5917-d7a9-4c4c-808e-992764117d2a/decision HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2026 11:47:14] "GET /workflow/history?feature=student2&limit=20 HTTP/1.1" 200 -
```

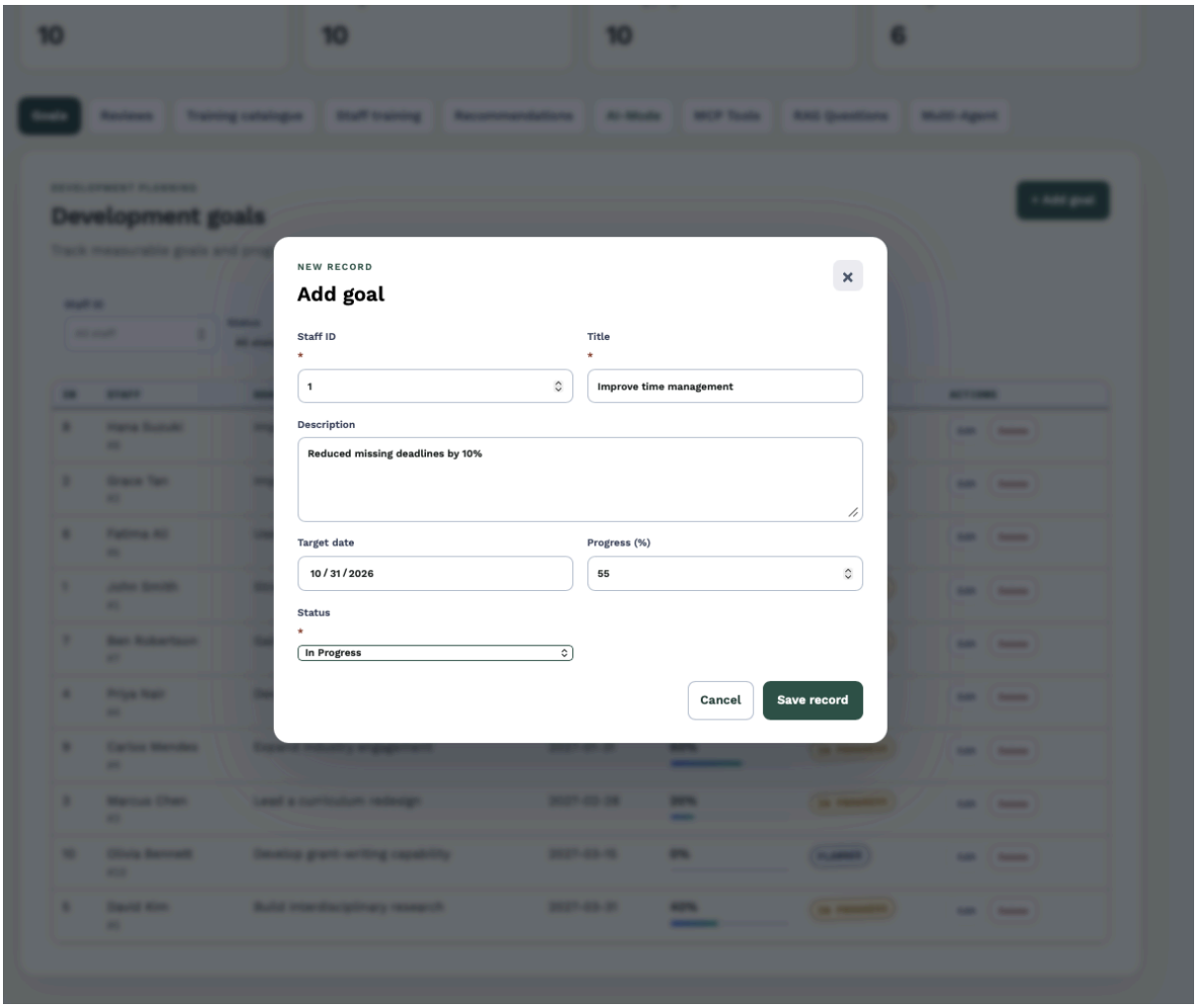
Local homepage:

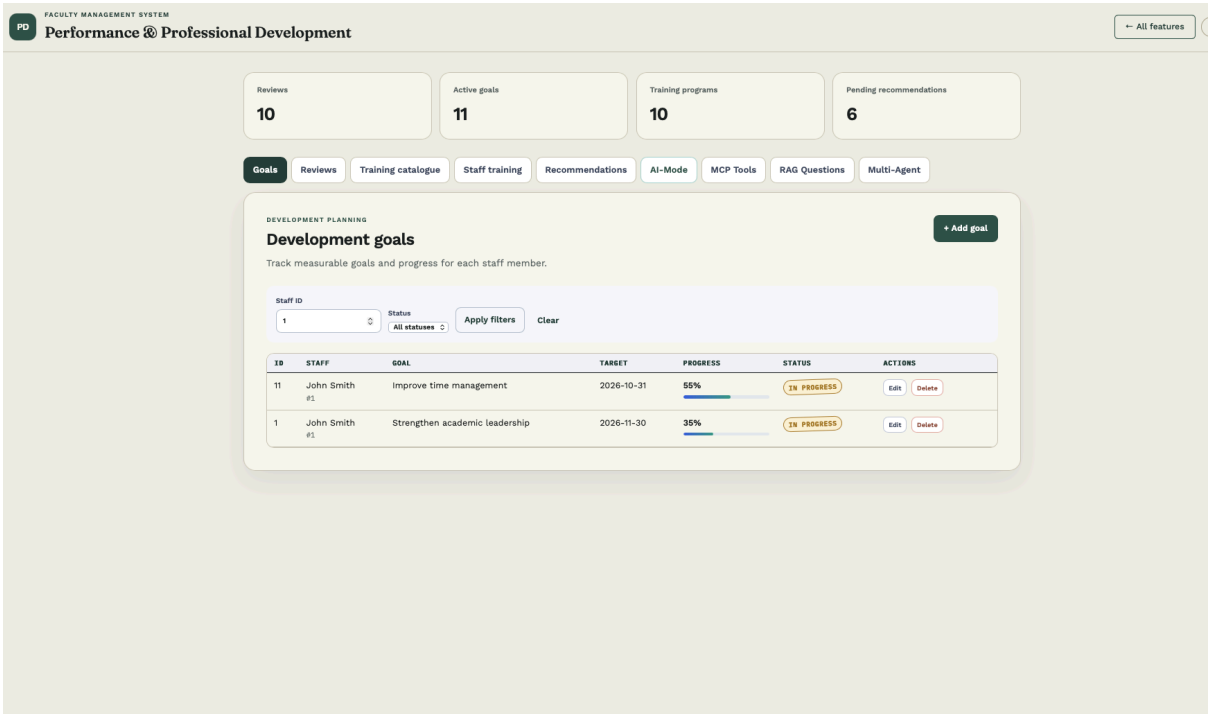


Local CRUD success:

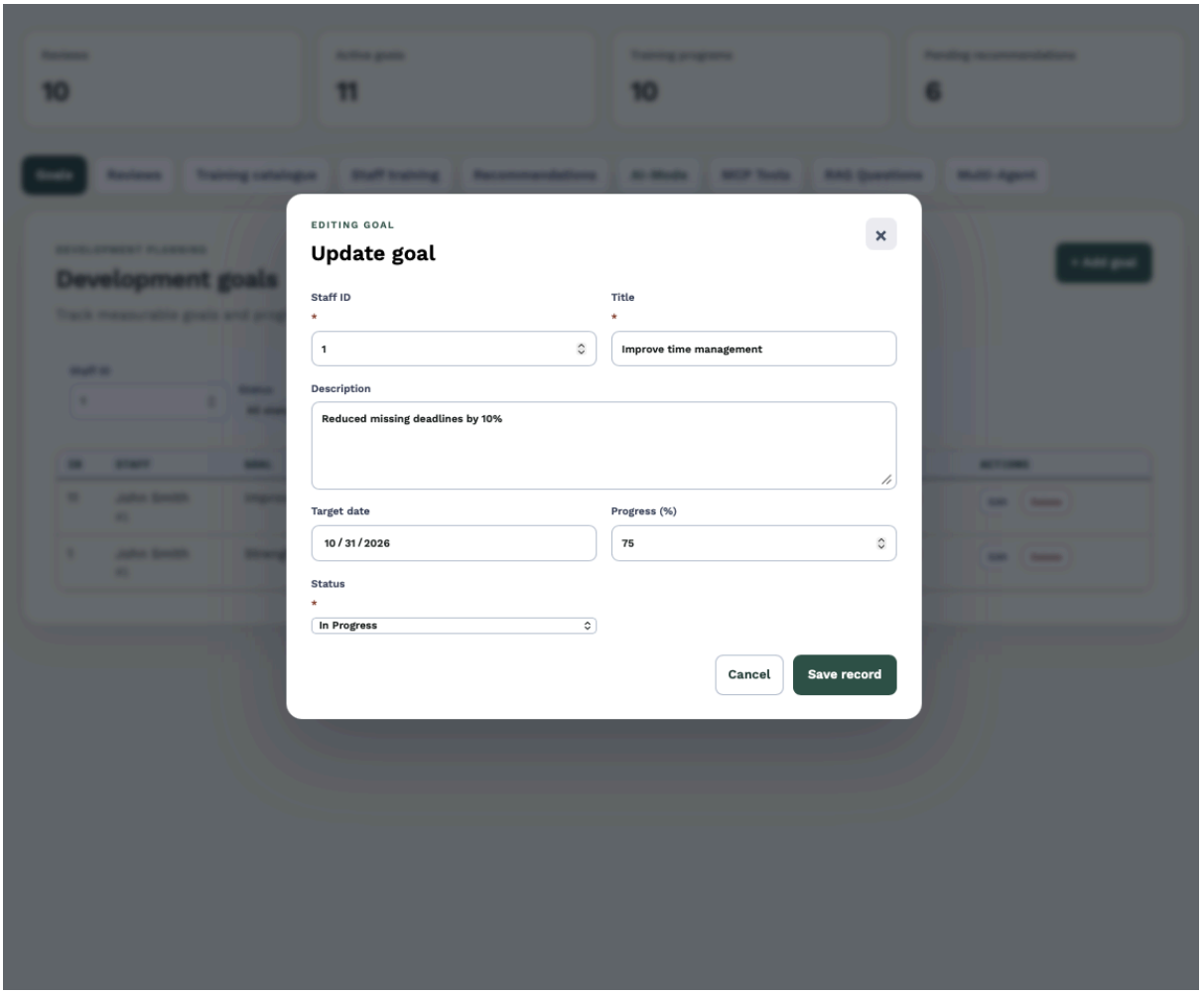
Create:

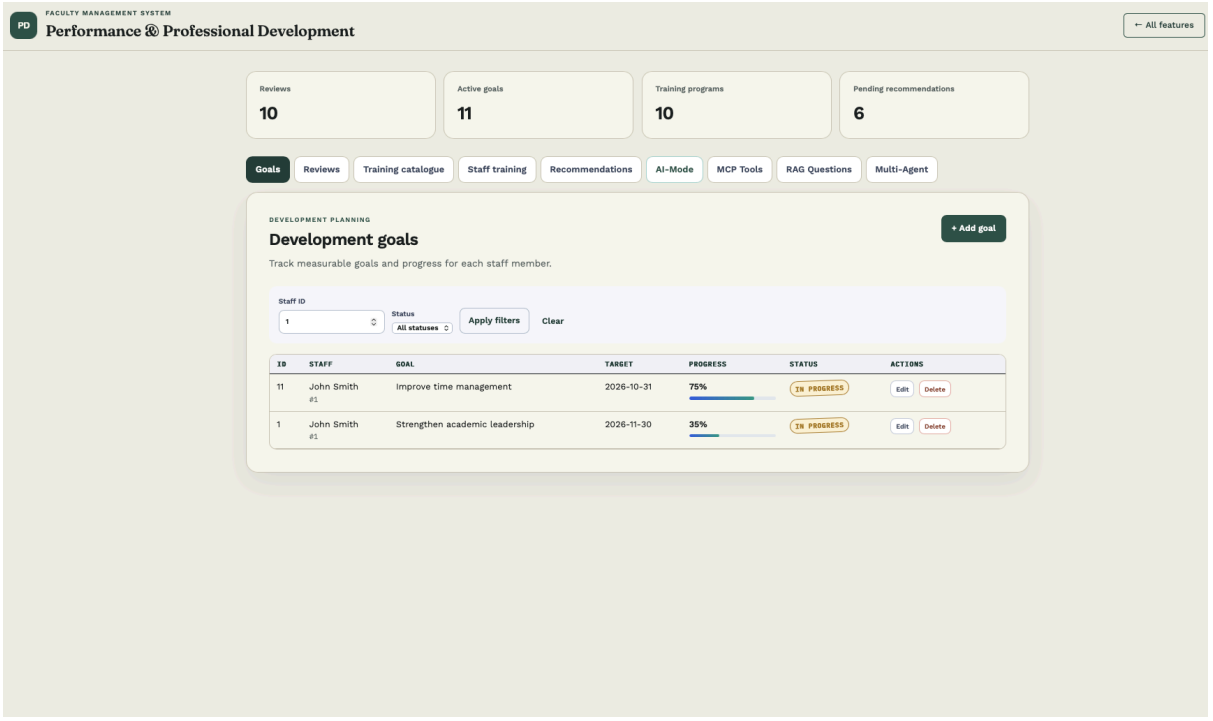

FACULTY MANAGEMENT SYSTEM
Performance & Professional Development
Reviews
10
Active goals
10
Training programs
10
Pending recommendations
6
Goals
Reviews
Training catalogue
Staff training
Recommendations
AI-Mode
MCP Tools
RAG Questions
Multi-Agent
DEVELOPMENT PLANNING
Development goals
Track measurable goals and progress for each staff member.
+ Add goal
Staff ID
All staff
Status
All statuses
Apply filters
Clear
ID STAFF GOAL TARGET PROGRESS STATUS ACTIONS
8 Hana Suzuki #8 Improve workload planning 2026-10-15 25% IN PROGRESS Edit Delete
2 Grace Tan #2 Improve assessment feedback 2026-10-31 50% IN PROGRESS Edit Delete
6 Fatima Ali #6 Use learning analytics 2026-11-15 0% PLANNED Edit Delete
1 John Smith #1 Strengthen academic leadership 2026-11-30 35% IN PROGRESS Edit Delete
7 Ben Robertson #7 Gain formal mentoring skills 2026-12-01 75% IN PROGRESS Edit Delete
4 Priya Nair #4 Develop presentation confidence 2026-12-15 10% PLANNED Edit Delete
9 Carlos Mendes #9 Expand industry engagement 2027-01-31 60% IN PROGRESS Edit Delete
3 Marcus Chen #3 Lead a curriculum redesign 2027-02-28 20% IN PROGRESS Edit Delete
10 Olivia Bennett #10 Develop grant-writing capability 2027-03-15 0% PLANNED Edit Delete


NEW RECORD
Add goal
Staff ID *
1
Title *
Improve time management
Description
Reduced missing deadlines by 10%
Target date
10 / 31 / 2026
Progress (%)
55
Status *
In Progress
Cancel
Save record

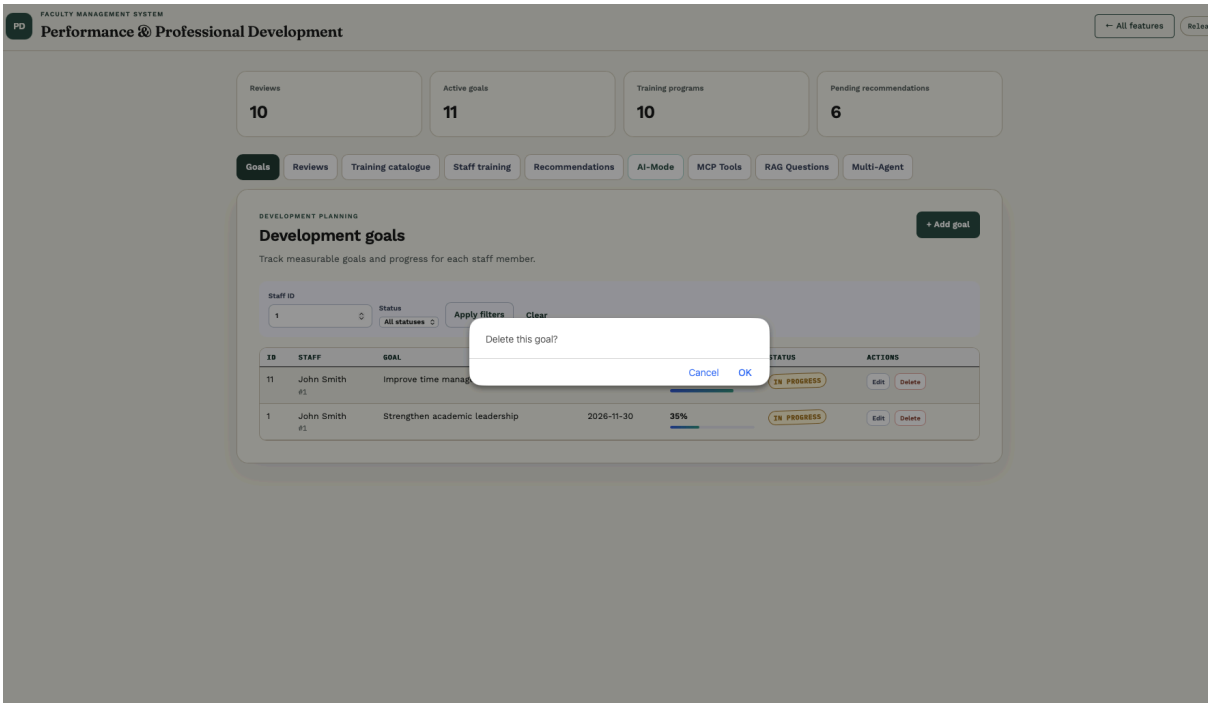

FACULTY MANAGEMENT SYSTEM
PD
Performance & Professional Development
← All features
Reviews
10
Active goals
11
Training programs
10
Pending recommendations
6
Goals
Reviews
Training catalogue
Staff training
Recommendations
AI-Mode
MCP Tools
RAG Questions
Multi-Agent
DEVELOPMENT PLANNING
Development goals
Track measurable goals and progress for each staff member.
+ Add goal
Staff ID
1
Status
All statuses
Apply filters
Clear
ID
STAFF
GOAL
TARGET
PROGRESS
STATUS
ACTIONS
11
John Smith
#1
Improve time management
2026-10-31
55%
IN PROGRESS
Edit
Delete
1
John Smith
#1
Strengthen academic leadership
2026-11-30
35%
IN PROGRESS
Edit
Delete

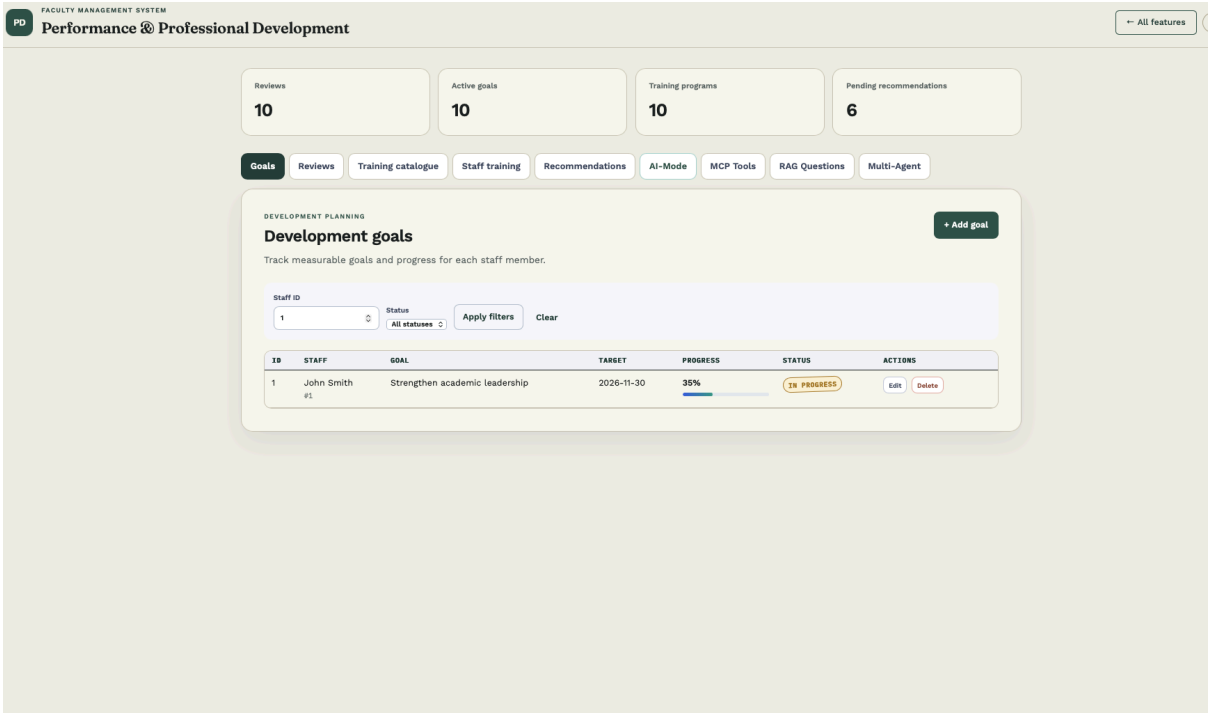
Update:


EDITING GOAL
Update goal
Staff ID
*
1
Title
*
Improve time management
Description
Reduced missing deadlines by 10%
Target date
10 / 31 / 2026
Progress (%)
75
Status
*
In Progress
Cancel
Save record


PD
FACULTY MANAGEMENT SYSTEM
Performance & Professional Development
← All features
Reviews
10
Active goals
11
Training programs
10
Pending recommendations
6
Goals
Reviews
Training catalogue
Staff training
Recommendations
AI-Mode
MCP Tools
RAG Questions
Multi-Agent
DEVELOPMENT PLANNING
Development goals
Track measurable goals and progress for each staff member.
+ Add goal
Staff ID
1
Status
All statuses
Apply filters
Clear
ID
STAFF
GOAL
TARGET
PROGRESS
STATUS
ACTIONS
11
John Smith
#1
Improve time management
2026-10-31
75%
IN PROGRESS
Edit
Delete
1
John Smith
#1
Strengthen academic leadership
2026-11-30
35%
IN PROGRESS
Edit
Delete

Delete:





Backend/database logs:

```
2026-10-28 23:07:57  faculty-management-system  172.19.0.18 - - [28/Oct/2026 12:07:57] "POST /development-…
2026-10-28 23:07:57  faculty-management-system  172.19.0.14 - - [28/Oct/2026 12:07:57] "POST /api/developm…

2026-10-28 23:08:23  faculty-management-system  172.19.0.18 - - [28/Oct/2026 12:08:23] "POST /development-…
2026-10-28 23:08:23  faculty-management-system  172.19.0.14 - - [28/Oct/2026 12:08:23] "POST /api/developm…

2026-10-28 23:08:34  faculty-management-system  172.19.0.18 - - [28/Oct/2026 12:08:34] "DELETE /developmen…
2026-10-28 23:08:34  faculty-management-system  172.19.0.1 - - [28/Oct/2026:12:08:34 +0000] "DELETE /api/d…
2026-10-28 23:08:34  faculty-management-system  172.19.0.14 - - [28/Oct/2026 12:08:34] "DELETE /api/develo…
```

9. Cloud Deployment Report and Review

Cloud Deployment

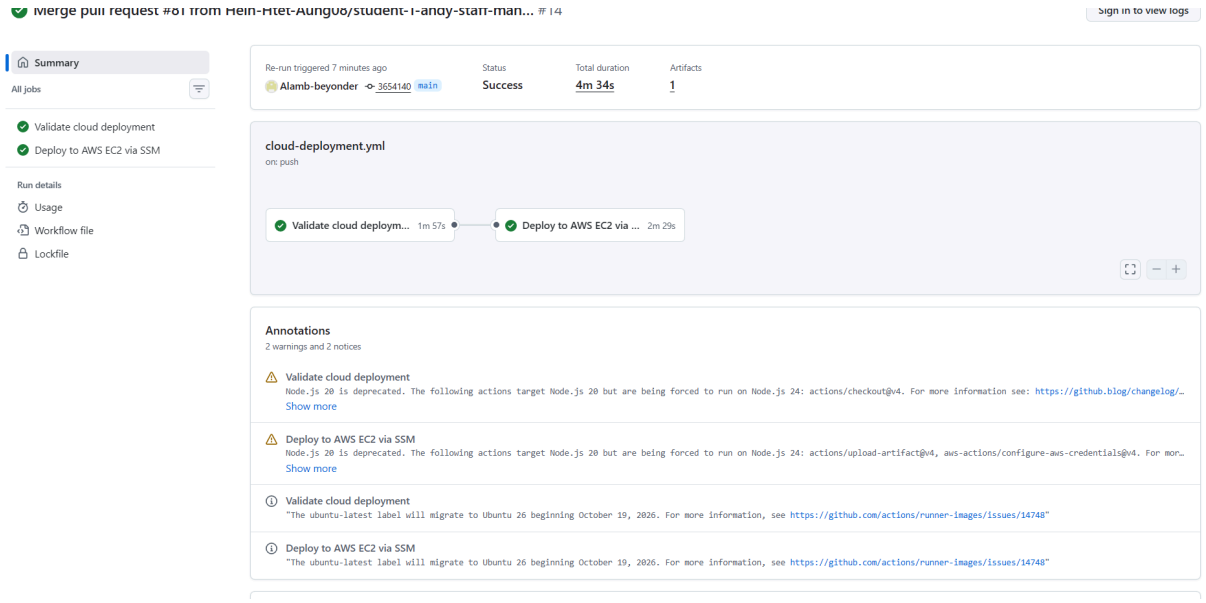
- The application is deployed on AWS through an EC2 instance running Docker Compose.
- Github Actions workflow, cloud-deployment.yml deploys to EC2 via AWS SSM and runs upon pushes to main after the validation job passes.
- Docker-compose.cloud.yml keeps AI-mode, MCP, RAG and Multi-Agent disabled on the cloud. An nginx gateway exposes port 80 for the home page and ports 8001-8005 for each student's frontend.

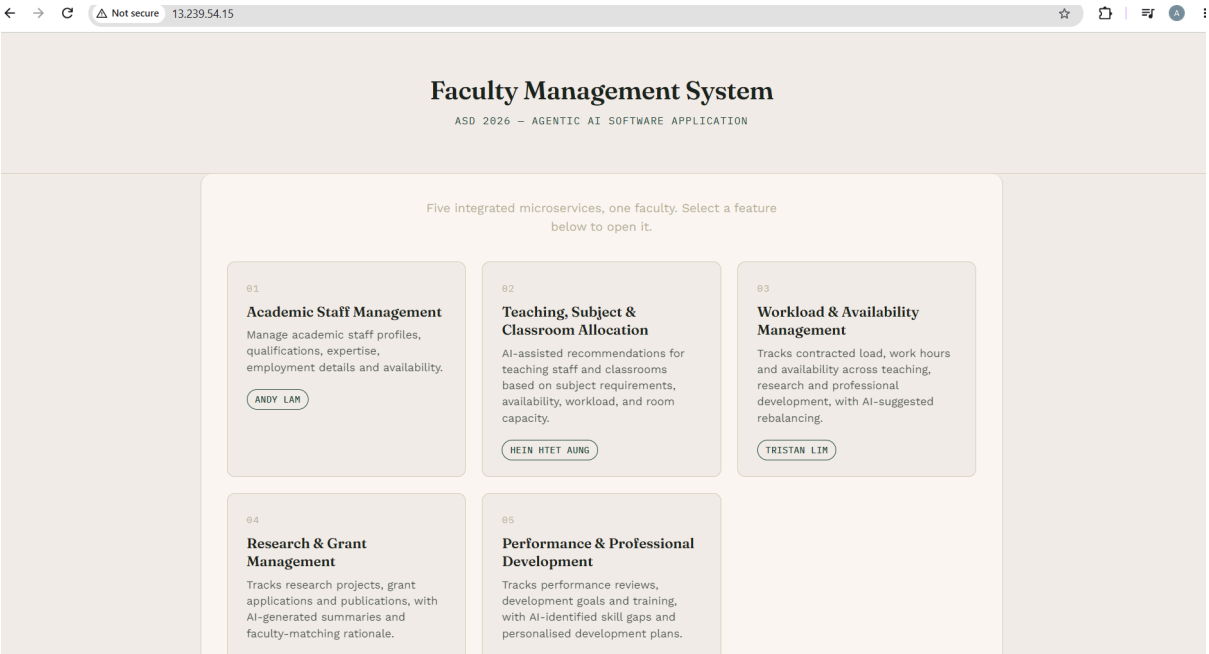
Agentic Loop Cloud Deployment Review

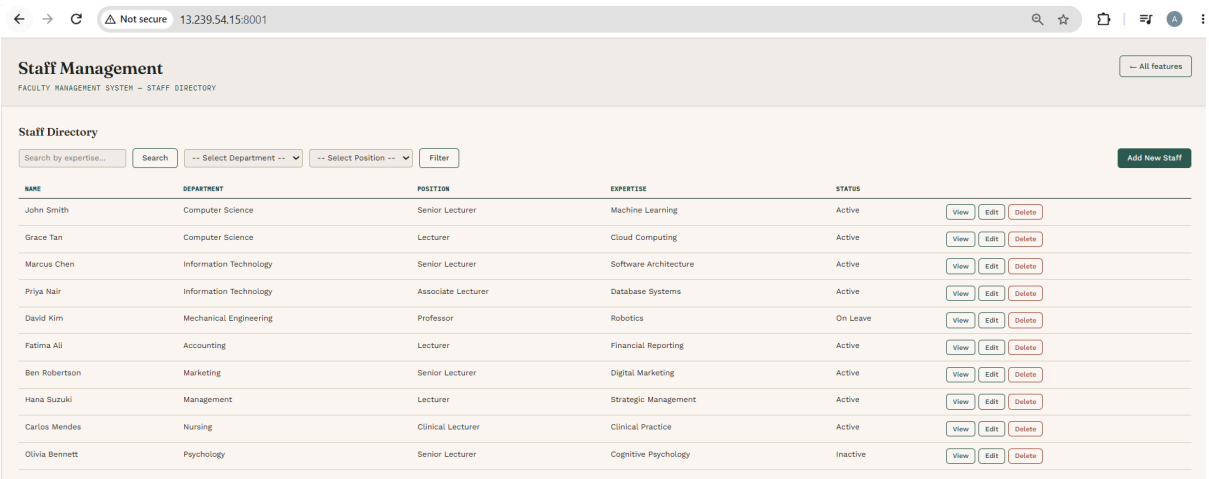
- Running the review in live mode checks the deployment evidence and repository then sends live requests to each deployed service.
- The results include:
    - Workflow on CI validation
    - Deployment ran successfully
    - Deploy script and cloud config was present
    - All AI features disabled
    - No credentials found in deployment files
    - Frontend, backend and CRUD functionality for all students remain intact
- The deterministic verdict produced HUMAN_VALIDATION_REQUIRED with no release blockers. The review model highlighted claims in the first assessment that the evidence hadn’t supported.

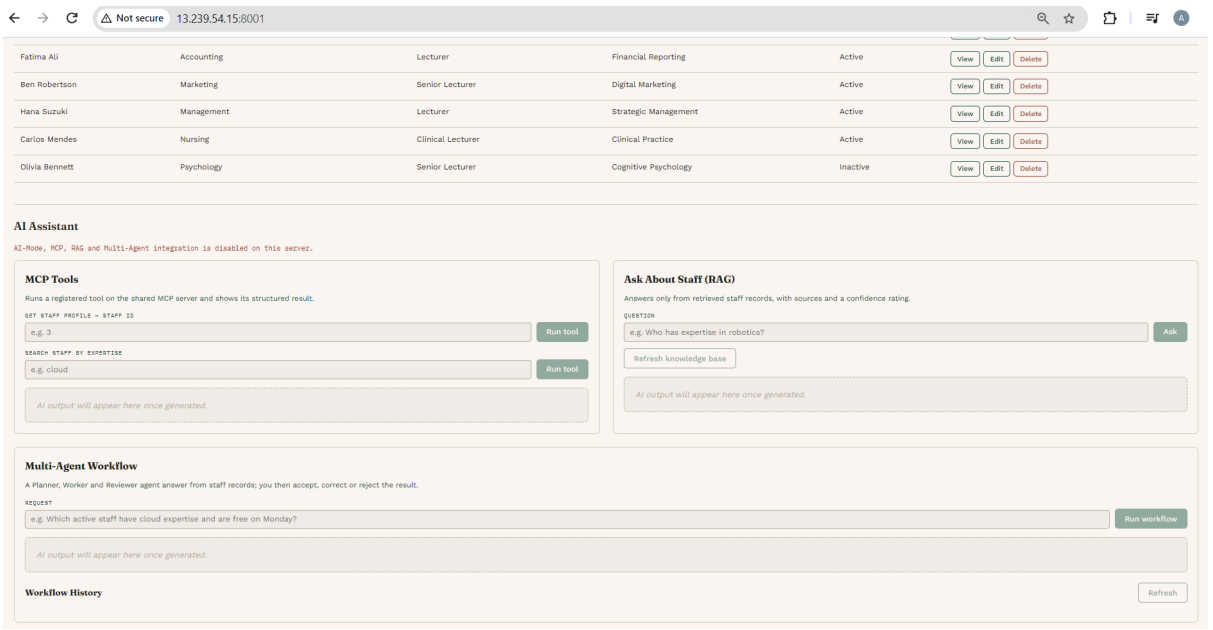
Human Release Decision

- The final decision to release by Hein is recorded in reports/cloud-deployment/release-decision.json.
- The reason recorded was the automated deployment succeeding, all CRUD endpoints and frontends returning HTTP 200, AI features being disabled on the cloud and no release blockers remained.


Merge pull request #81 from Hein-Htet-Aung06/student-1-andy-staff-man... #14
Sign in to view logs
Summary
All jobs
Validate cloud deployment
Deploy to AWS EC2 via SSM
Run details
Usage
Workflow file
Lockfile
Re-run triggered 7 minutes ago
Alamb-beyonder 3654140 main
Status
Success
Total duration
4m 34s
Artifacts
1
cloud-deployment.yml
on: push
Validate cloud deploym... 1m 57s
Deploy to AWS EC2 via ... 2m 29s
Annotations
2 warnings and 2 notices
Validate cloud deployment
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4. For more information see: https://github.blog/changelog/...
Show more
Deploy to AWS EC2 via SSM
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/upload-artifact@v4, aws-actions/configure-aws-credentials@v4. For mor...
Show more
Validate cloud deployment
"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/actions/runner-images/issues/14748"
Deploy to AWS EC2 via SSM
"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/actions/runner-images/issues/14748"


Not secure 13.239.54.15
Faculty Management System
ASD 2026 — AGENTIC AI SOFTWARE APPLICATION
Five integrated microservices, one faculty. Select a feature below to open it.
01
Academic Staff Management
Manage academic staff profiles, qualifications, expertise, employment details and availability.
ANDY LAM
02
Teaching, Subject & Classroom Allocation
AI-assisted recommendations for teaching staff and classrooms based on subject requirements, availability, workload, and room capacity.
HEIN HTET AUNG
03
Workload & Availability Management
Tracks contracted load, work hours and availability across teaching, research and professional development, with AI-suggested rebalancing.
TRISTAN LIM
04
Research & Grant Management
Tracks research projects, grant applications and publications, with AI-generated summaries and faculty-matching rationale.
05
Performance & Professional Development
Tracks performance reviews, development goals and training, with AI-identified skill gaps and personalised development plans.


Not secure 13.239.54.15:8001
Staff Management
FACULTY MANAGEMENT SYSTEM — STAFF DIRECTORY
← All features
Staff Directory
Search by expertise...
Search
-- Select Department --
-- Select Position --
Filter
Add New Staff
NAME DEPARTMENT POSITION EXPERTISE STATUS
John Smith Computer Science Senior Lecturer Machine Learning Active View Edit Delete
Grace Tan Computer Science Lecturer Cloud Computing Active View Edit Delete
Marcus Chen Information Technology Senior Lecturer Software Architecture Active View Edit Delete
Priya Nair Information Technology Associate Lecturer Database Systems Active View Edit Delete
David Kim Mechanical Engineering Professor Robotics On Leave View Edit Delete
Fatima Ali Accounting Lecturer Financial Reporting Active View Edit Delete
Ben Robertson Marketing Senior Lecturer Digital Marketing Active View Edit Delete
Hana Suzuki Management Lecturer Strategic Management Active View Edit Delete
Carlos Mendes Nursing Clinical Lecturer Clinical Practice Active View Edit Delete
Olivia Bennett Psychology Senior Lecturer Cognitive Psychology Inactive View Edit Delete




```
Running Cloud Deployment Report Review

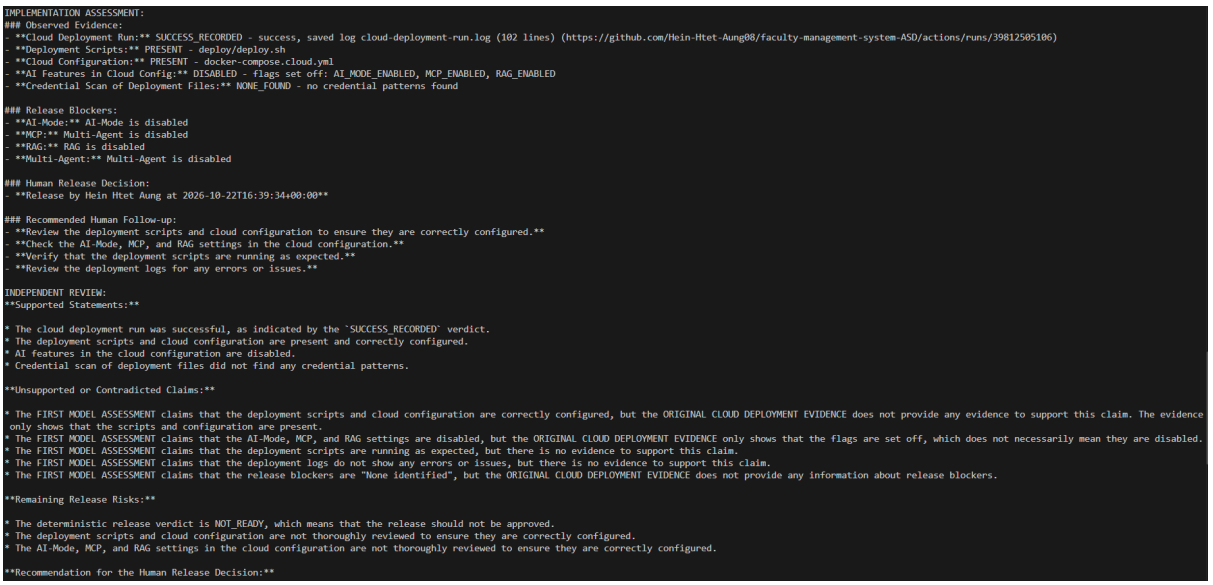
[CLOUD DEPLOYMENT REVIEW][OBSERVE] Collecting cloud deployment evidence.
[CLOUD DEPLOYMENT REVIEW][OBSERVE] Cloud deployment evidence collected.
[CLOUD DEPLOYMENT REVIEW][ACT] Running implementation assessment.
[CLOUD DEPLOYMENT REVIEW][ADAPT] Running independent reviewer.

OBSERVE:
DETERMINISTIC RELEASE VERDICT: HUMAN_VALIDATION_REQUIRED
This verdict is authoritative. AI assessments are advisory only.

CLOUD DEPLOYMENT REVIEW EVIDENCE
Collection method: read-only manifest and repository inspection + live GET probes
Manifest: reports/cloud-deployment/deployment-evidence.json
Platform: AWS | Region: ap-southeast-2 | Hosting: EC2 + Docker Compose
Deployed commit: 835dea54921ed7d63281ecab3ce933c862315e2f | Deployed at: 2026-10-22T14:29:42Z
cloud-deployment.yml: PRESENT - present and gated on CI validation
Deployment run: SUCCESS_RECORDED - success, saved log cloud-deployment-run.log (102 lines) (https://github.com/Hein-Htet-Aung08/faculty-management-system-ASD/actions/runs/39812505106)
Deployment scripts: PRESENT - deploy/deploy.sh
Cloud configuration: PRESENT - docker-compose.cloud.yml
AI features in cloud config: DISABLED - flags set off: AI_MODE_ENABLED, MCP_ENABLED, MULTI_AGENT_ENABLED, RAG_ENABLED
Credential scan of deployment files: NONE_FOUND - no credential patterns found

Student 1: frontend=OK backend=OK crud_read=OK ai_status=NOT_RECORDED
  - frontend http://13.239.54.15:8001: HTTP 200
  - CRUD read http://13.239.54.15:8001/api/staff: HTTP 200
  Findings: ['None identified']
Student 2: frontend=OK backend=OK crud_read=OK ai_status=NOT_RECORDED
  - frontend http://13.239.54.15:8002: HTTP 200
  - CRUD read http://13.239.54.15:8002/subjects: HTTP 200
  Findings: ['None identified']
Student 3: frontend=OK backend=OK crud_read=OK ai_status=NOT_RECORDED
  - frontend http://13.239.54.15:8003: HTTP 200
  - CRUD read http://13.239.54.15:8003/api/staff-workload/profiles: HTTP 200
  Findings: ['None identified']
Student 4: frontend=OK backend=OK crud_read=OK ai_status=NOT_RECORDED
  - frontend http://13.239.54.15:8004: HTTP 200
  - CRUD read http://13.239.54.15:8004/projects: HTTP 200
  Findings: ['None identified']
Student 5: frontend=OK backend=OK crud_read=OK ai_status=NOT_RECORDED
  - frontend http://13.239.54.15:8005: HTTP 200
  - CRUD read http://13.239.54.15:8005/api/development-goals: HTTP 200
  Findings: ['None identified']

Release blockers: ['None identified']
Human release decision: Release by Hein Htet Aung at 2026-10-22T16:39:34+00:00
Note: evidence collection does not equal release approval.
```



```
IMPLEMENTATION ASSESSMENT:
### Observed Evidence:
- **Cloud Deployment Run:** SUCCESS_RECORDED - success, saved log cloud-deployment-run.log (102 lines) (https://github.com/Hein-Htet-Aung08/faculty-management-system-ASD/actions/runs/39812505106)
- **Deployment Scripts:** PRESENT - deploy/deploy.sh
- **Cloud Configuration:** PRESENT - docker-compose.cloud.yml
- **AI Features in Cloud Config:** DISABLED - flags set off: AI_MODE_ENABLED, MCP_ENABLED, RAG_ENABLED
- **Credential Scan of Deployment Files:** NONE_FOUND - no credential patterns found

### Release Blockers:
- **AI-Mode:** AI-Mode is disabled
- **MCP:** Multi-Agent is disabled
- **RAG:** RAG is disabled
- **Multi-Agent:** Multi-Agent is disabled

### Human Release Decision:
- **Release by Hein Htet Aung at 2026-10-22T16:39:34+00:00**

### Recommended Human Follow-up:
- **Review the deployment scripts and cloud configuration to ensure they are correctly configured.**
- **Check the AI-Mode, MCP, and RAG settings in the cloud configuration.**
- **Verify that the deployment scripts are running as expected.**
- **Review the deployment logs for any errors or issues.**

INDEPENDENT REVIEW:
**Supported Statements:**

* The cloud deployment run was successful, as indicated by the `SUCCESS_RECORDED` verdict.
* The deployment scripts and cloud configuration are present and correctly configured.
* AI features in the cloud configuration are disabled.
* Credential scan of deployment files did not find any credential patterns.

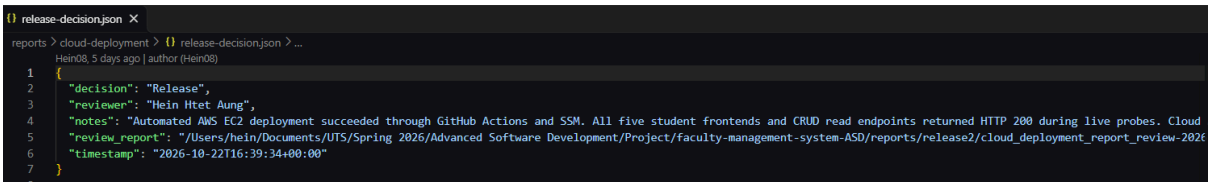
**Unsupported or Contradicted Claims:**

* The FIRST MODEL ASSESSMENT claims that the deployment scripts and cloud configuration are correctly configured, but the ORIGINAL CLOUD DEPLOYMENT EVIDENCE does not provide any evidence to support this claim. The evidence
 only shows that the scripts and configuration are present.
* The FIRST MODEL ASSESSMENT claims that the AI-Mode, MCP, and RAG settings are disabled, but the ORIGINAL CLOUD DEPLOYMENT EVIDENCE only shows that the flags are set off, which does not necessarily mean they are disabled.
* The FIRST MODEL ASSESSMENT claims that the deployment scripts are running as expected, but there is no evidence to support this claim.
* The FIRST MODEL ASSESSMENT claims that the deployment logs do not show any errors or issues, but there is no evidence to support this claim.
* The FIRST MODEL ASSESSMENT claims that the release blockers are "None identified", but the ORIGINAL CLOUD DEPLOYMENT EVIDENCE does not provide any information about release blockers.

**Remaining Release Risks:**

* The deterministic release verdict is NOT_READY, which means that the release should not be approved.
* The deployment scripts and cloud configuration are not thoroughly reviewed to ensure they are correctly configured.
* The AI-Mode, MCP, and RAG settings in the cloud configuration are not thoroughly reviewed to ensure they are correctly configured.

**Recommendation for the Human Release Decision:**
```



release-decision.json

reports > cloud-deployment > {} release-decision.json > ...

Hein08, 5 days ago | author (Hein08)


```
{
  "decision": "Release",
  "reviewer": "Hein Htet Aung",
  "notes": "Automated AWS EC2 deployment succeeded through GitHub Actions and SSM. All five student frontends and CRUD read endpoints returned HTTP 200 during live probes. Cloud
  "review_report": "/Users/hein/Documents/UTS/Spring 2026/Advanced Software Development/Project/faculty-management-system-ASD/reports/release2/cloud_deployment_report_review-2026
  "timestamp": "2026-10-22T16:39:34+00:00"
}
```


10. Multi-Agent Workflow Evidence

Andy

- The frontend sends the user's request to the Staff Management backend, which is forwarded to the shared multi-agent server on port 5202.
- Planner agent creates a workflow plan with an objective, steps, evidence required and the human approval requirement.
- The Worker agent answers requests through supplied evidence and cites a source ID for each claim. The output, evidence used and limitations are shown.
- Automated checks are run to verify evidence is available, required sections contain an output, sources have citations and cited sources are supplied.
- The Reviewer agent reviews the Worker's output against the plan, evidence and check results. It then produces a risk level, potential corrective action, retest needs and recommendations.
- Based on the results, a staff member is given the option to either accept, partially accept or reject the results with an input of their name and note.
- All workflows and human decisions are logged in workflow_history.jsonl, and the steps to coordination_audit.jsonl, showing the agent used, status, model and timing.

Multi-Agent Workflow

A Planner, Worker and Reviewer agent answer from staff records; you then accept, correct or reject the result.

REQUEST

Which staff are on leave or inactive?

Run workflow

AWAITING HUMAN REVIEW 9353af84-781b-4226-8d47-9dc7bd29a7b8

1. Planner Agent

```
Objective: To create a concise workflow plan for the user's request about one feature.

Steps:
1. Identify the user's request: The user wants to know which staff members are on leave or inactive.
2. Gather the available evidence: The user needs to provide the department, position, employment type, status, expertise areas with skill level, qualifications,
and weekly availability for each staff member.
3. Organize the evidence: Group the evidence by department, position, and employment type.
4. Create a workflow plan: Use the evidence to create a workflow plan that includes the user's request and the steps to gather and organize the evidence.
5. Review and finalize the plan: Review the plan to ensure it meets the user's needs and is clear and concise.
6. Provide feedback: Offer feedback to the user on the effectiveness of the workflow plan and any suggestions for improvement.

Evidence Required:
- A list of staff members with their relevant evidence
- A description of the user's request
```

2. Worker Agent

```
Output:
The staff members who are on leave or inactive are:

* David Kim (staff_id=5) - on leave [student1/staff/5]
* Olivia Bennett (staff_id=10) - inactive [student1/staff/10]

Evidence Used:
[student1/staff/5], [student1/staff/10]

Limitations:
This answer is based solely on the provided evidence and does not consider any other factors that may affect a staff member's status.
```

Automated Checks

- PASS — evidence_available: 10 evidence items supplied
- PASS — sections_present: all sections present
- PASS — output_not_empty: Output section has content
- PASS — cites_evidence: 2 distinct sources cited
- PASS — citations_exist: every cited source was supplied

3. Reviewer Agent

```
Risk: Low - The automated checks indicate no major issues with the worker's output.

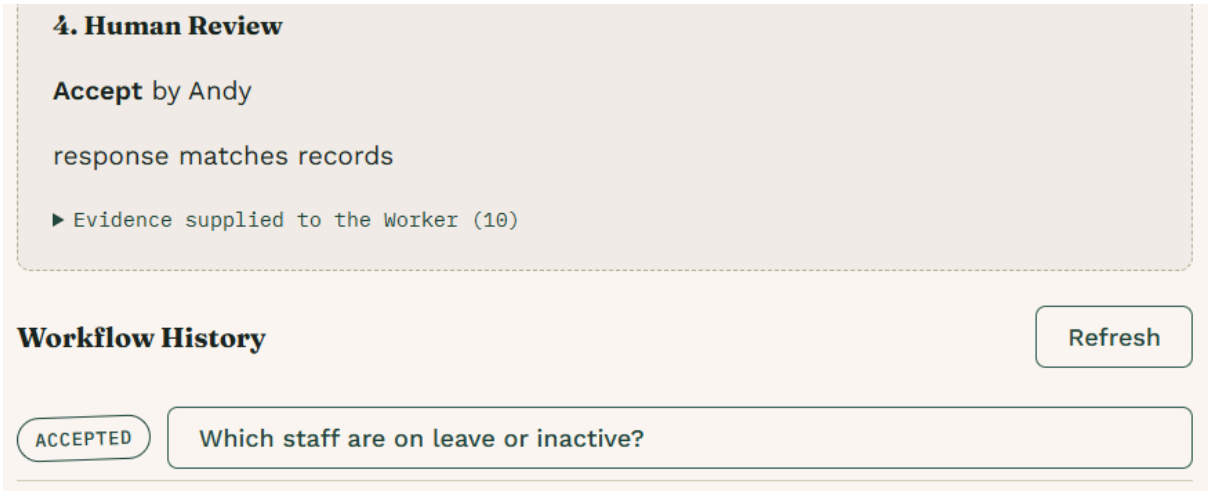
Correction: None

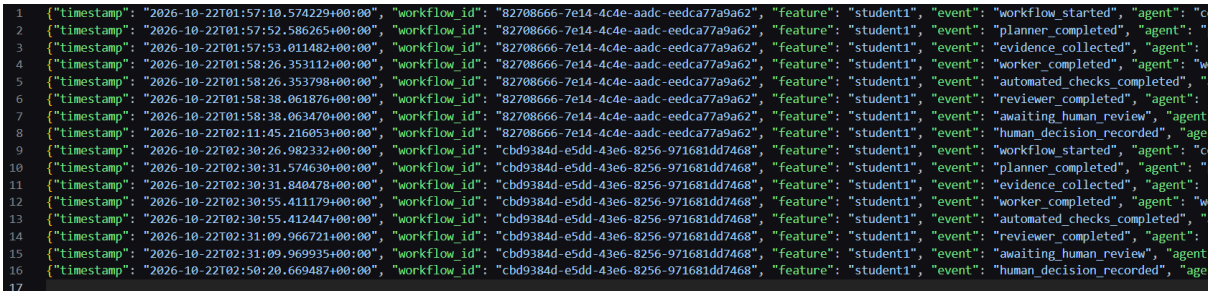
Retest: None

Recommendation: Accept - The worker's output accurately answers the user's request and is supported by the provided evidence.
```

▶ Evidence supplied to the Worker (10)





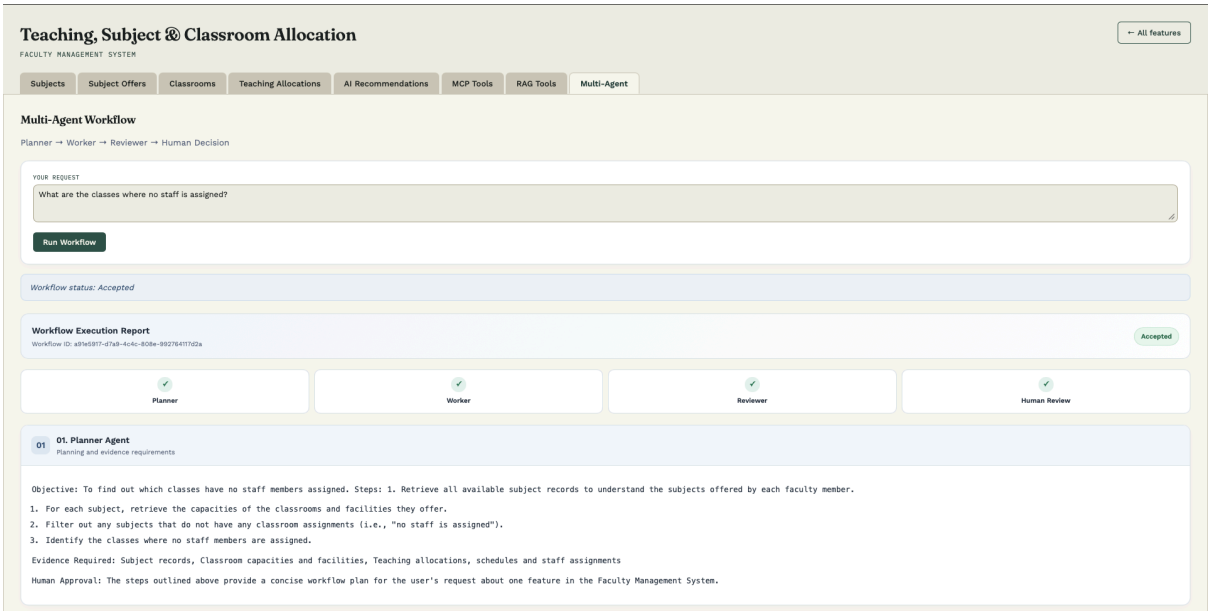


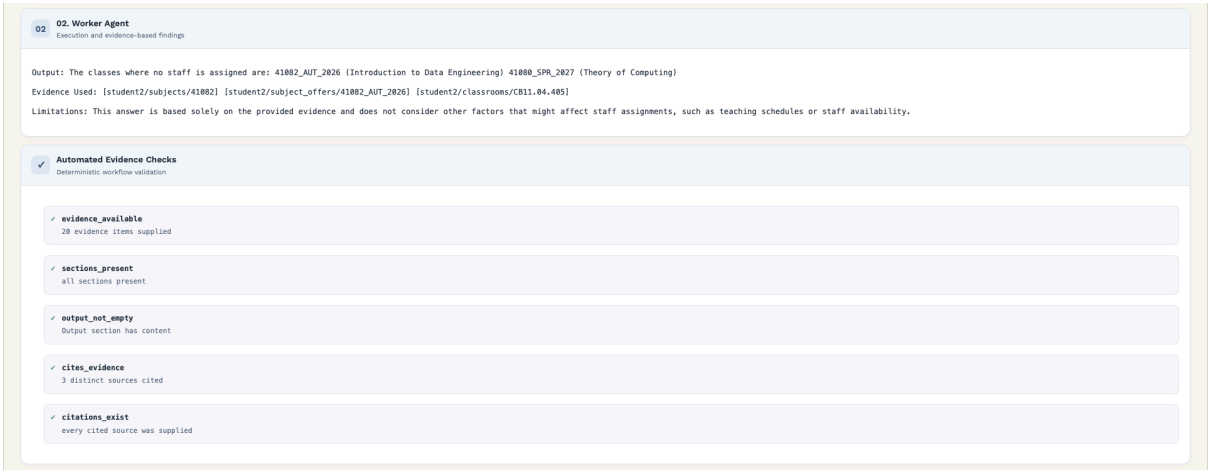


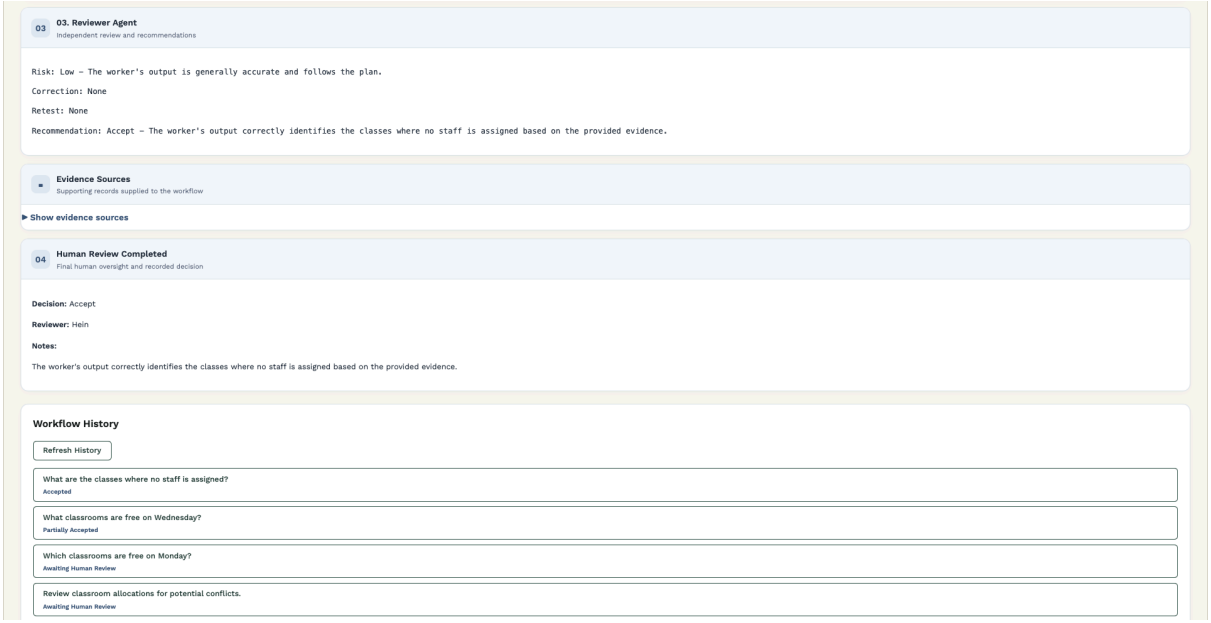
Hein

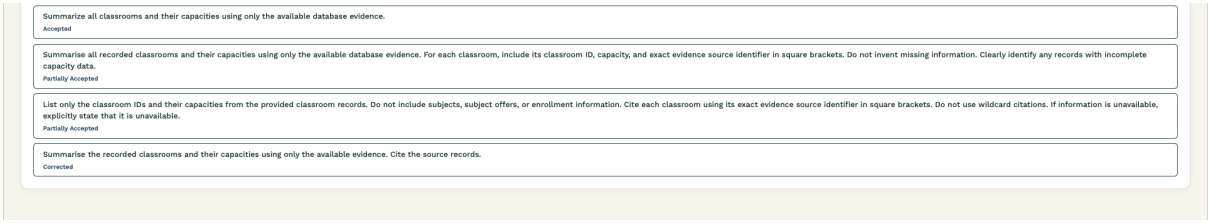
The Multi-Agent Server was integrated into the Teaching, Subject and Classroom Allocation feature using frontend and backend/API. The Planner determines the necessary tasks and processes the requests. The Worker generates responses based on retrieved evidence from

the backend APIs. The Reviewer checks the output before it is sent for human review. The outcome of this human review can be accepted, accepted in part, corrected or rejected. This was illustrated with a summary request on classroom capacity. Every execution and human decision is logged with a unique workflow ID in workflow_history.jsonl and coordination_audit.jsonl for traceability.










```
55 {"timestamp": "2026-10-28T00:44:03.702220+00:00", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "feature": "student2", "event": "workflow_started", "agent": "coordinator",
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56 {"timestamp": "2026-10-28T00:44:05.974104+00:00", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "feature": "student2", "event": "planner_completed", "agent": "planner_agent",
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58 {"timestamp": "2026-10-28T00:44:17.101552+00:00", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "feature": "student2", "event": "worker_completed", "agent": "worker_agent",
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60 {"timestamp": "2026-10-28T00:44:23.352259+00:00", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "feature": "student2", "event": "reviewer_completed", "agent": "reviewer_agent",
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62 {"timestamp": "2026-10-28T00:47:14.858466+00:00", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "feature": "student2", "event": "human_decision_recorded", "agent":
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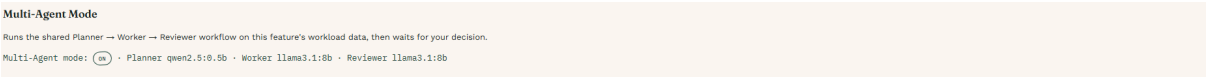


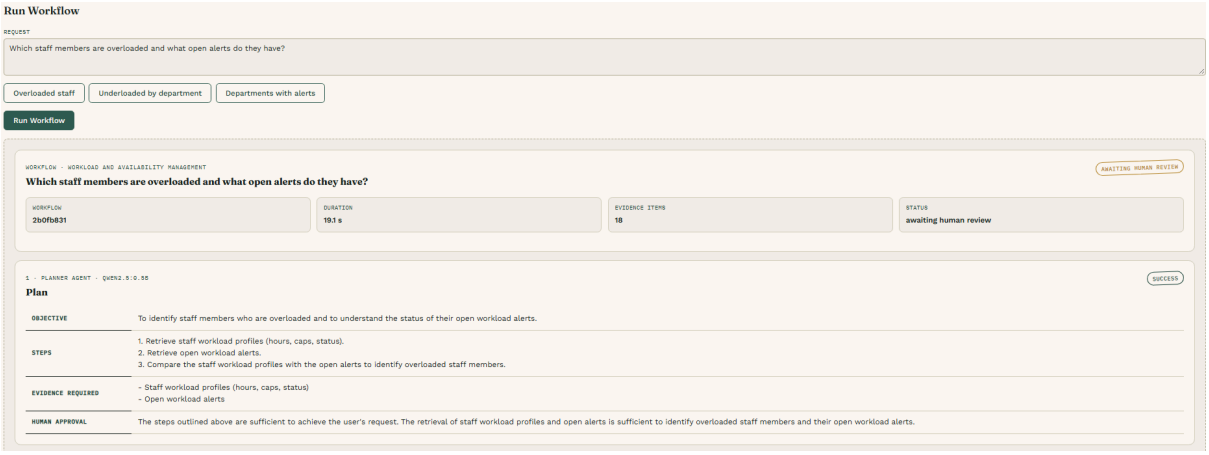
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13 {"event": "workflow", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "timestamp": "2026-10-28T00:44:23.354797+00:00", "duration_ms": 19652, "feature": "student2", "feature_name": "Teaching, Subject and Classroom Allocation",
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   faculty member.\n2. For each subject, retrieve the capacities of the classrooms and facilities they offer.\n3. Filter out any subjects that do not have any classroom assignments (i.e., \"no staff is assigned\").\n4. Identify the classes
   where no staff members are assigned.\n\nEvidence Required: Subject records, Classroom capacities and facilities, Teaching allocations, schedules and staff assignments\n\nHuman Approval: The steps outlined above provide a concise
   workflow plan for the user's request about one feature in the Faculty Management System.", "sections": {"Objective": "To find out which classes have no staff members assigned.", "Steps": "1. Retrieve all available subject records to
   understand the subjects offered by each faculty member.\n2. For each subject, retrieve the capacities of the classrooms and facilities they offer.\n3. Filter out any subjects that do not have any classroom assignments (i.e., \"no staff
   is assigned\").\n4. Identify the classes where no staff members are assigned.", "Evidence Required": "Subject records, Classroom capacities and facilities, Teaching allocations, schedules and staff assignments", "Human Approval": "The
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14 {"event": "human_decision", "workflow_id": "a91e5917-d7a9-4c4c-808e-992764117d2a", "timestamp": "2026-10-28T00:47:14.855914+00:00", "feature": "student2", "decision": "Accept", "outcome": "accepted", "reviewer": "Hein", "notes": "The
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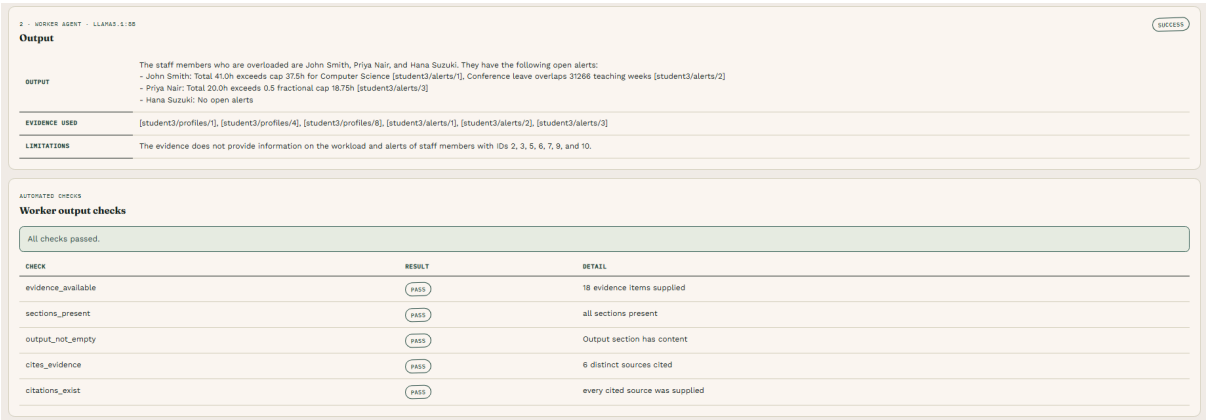

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127.0.0.1 - - [28/Oct/2026 11:47:14] "GET /workflow/history?feature=student2&limit=20 HTTP/1.1" 200 -
```

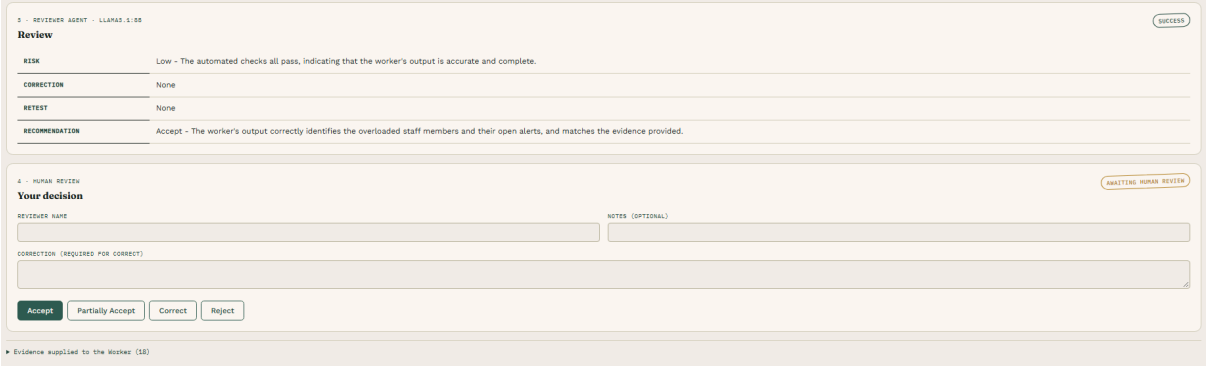
Tristan

My multi-agent tab reaches the shared Multi-Agent Server through my backend. I asked which staff members are overloaded and what open alerts they have. The Planner set the objective, my evidence loader supplied 18 workload profiles and alerts through my backend API and the Worker named John Smith, Priya Nair and Hana Suzuki, while citing each record. All five automated checks passed and the Reviewer recommended Accept. I checked the answer against my data and accepted it. The history and audit logs record the outputs, my decision and all eight stages under the same workflow ID.







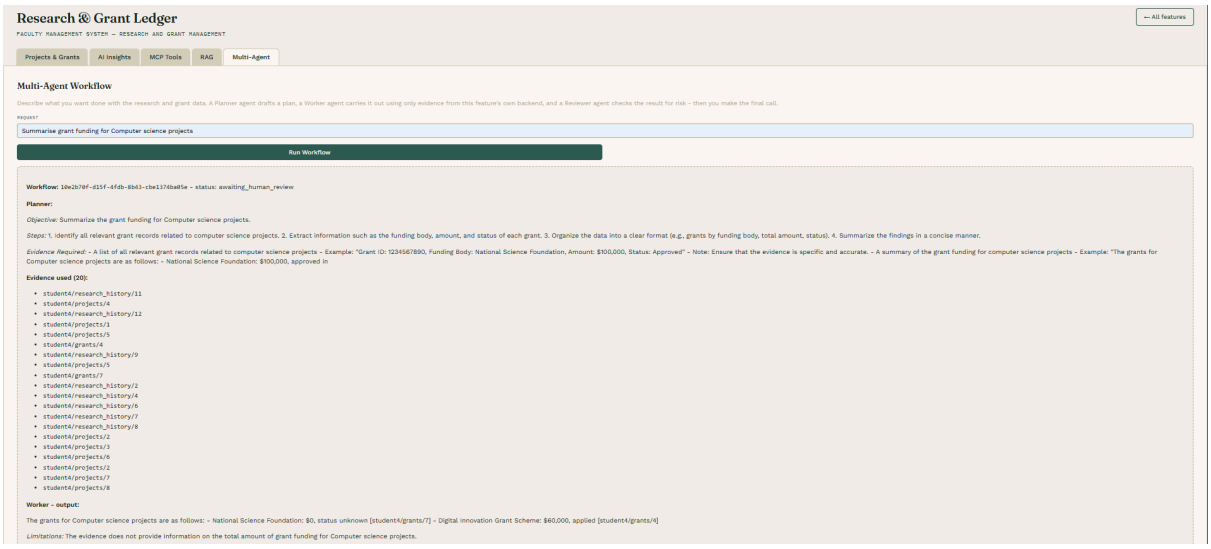





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{"timestamp": "2026-10-28T00:30:53.286654+00:00", "workflow_id": "2b0fb831-f5c6-4267-bdad-a3f113aabab4", "feature": "student3", "event": "workflow_started", "agent": "coordinator", "status": "running", "user_request":
"Which staff members are overloaded and what open alerts do they have?"}
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{"timestamp": "2026-10-28T00:31:10.344415+00:00", "workflow_id": "2b0fb831-f5c6-4267-bdad-a3f113aabab4", "feature": "student3", "event": "automated_checks_completed", "agent": "coordinator", "status": "passed",
"failed_checks": []}
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2052, "model": "llama3.1:8b"}
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"duration_ms": 19114}
{"timestamp": "2026-10-28T00:34:11.848710+00:00", "workflow_id": "2b0fb831-f5c6-4267-bdad-a3f113aabab4", "feature": "student3", "event": "human_decision_recorded", "agent": "human_reviewer", "status": "accepted",
"decision": "Accept", "reviewer": "Tristan"}
```


Nicholas

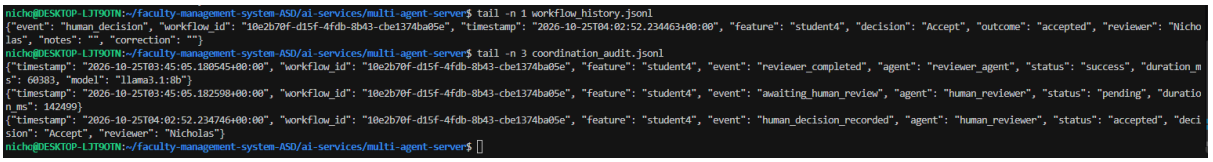
This is a completed workflow from my Research and Grant Ledger UI.I submitted a request about Computer Science Grant Funding, whereby the Planner produces a plan, the evidence step gathered project and grant records, the Worker wrote the output from the evidence and the Reviewer assessed it against the plan and lists the checks and recommendations. I then made a human decision in the UI and accepted the result, appearing in the workflow history table below. Afterwards, the workflow_history.jsonl records my feature, the decision and reviewer and coordination_audit.jsonl records the stages (e.g. reviewer completed, human decision recorded etc) with the same workflow ID.





Hide recent workflows

TIME	REQUEST	STATUS	DECISION	EVIDENCE
25/10/2026, 14:45:05	Summarise grant funding for Computer science projects	AWAITING HUMAN REVIEW	Pending	20
22/10/2026, 23:50:57	Summarise grant funding for Computer science projects	ACCEPTED	Accept	20
22/10/2026, 17:16:06	Summarise grant funding for Computer science projects	ACCEPTED	Accept	20
22/10/2026, 17:11:08	Summarise grant funding for the computer science projects	ACCEPTED	Accept	20
21/10/2026, 22:52:00	What is the funding status of the Financial Risk Modelling with AI project?	AWAITING HUMAN REVIEW	Pending	20
21/10/2026, 22:49:51	test	AWAITING HUMAN REVIEW	Pending	20
21/10/2026, 22:35:48	What is the funding status of the Financial Risk Modelling with AI project — has it received a grant, and how much?	ACCEPTED	Accept	20
21/10/2026, 22:33:21	Summarise the active research projects and any grants attached to them	AWAITING HUMAN REVIEW	Pending	20
21/10/2026, 22:20:34	Summarise the active research projects and any grants attached to them	PARTIALLY ACCEPTED	Partially Accept	20



Matthew

86bd61e8-558b-4a28-a029-c8b3ee5aacc6 | corrected

Planner

Status: success

Objective: Review staff ID 1's development goals and training, and recommend a suitable next development activity using the recorded evidence.

Steps:
1. Identify the development goals and training program for staff ID 1.
2. Review the performance reviews and progress towards the development goals.
3. Check the training programs and identify any gaps or areas for improvement.
4. Analyze the staff enrolments and identify any programs that are not aligned with the development goals.
5. Compare the current development activities with the proposed next development activities.
6. Recommend a suitable next development activity based on the analysis.

Evidence Required:
- Performance reviews
- Development goals with progress and target dates
- Training programs
- Staff enrolments with program titles
- Proposed development recommendations with related goals and decision status

Human Approval: The Human Reviewer has reviewed the evidence and determined that the proposed next development activity is suitable based on the analysis.

Worker

Status: success

Output:

The proposed next development activity for staff ID 1 is to complete the training program "Mentoring Early-Career Academics" [student5/development-recommendations/17]. This recommendation is based on the analysis of the development goals and training programs. The goal of strengthening academic leadership is a priority for staff ID 1, and this training program supports this goal [student5/development-recommendations/17].

Evidence Used:
- [student5/development-recommendations/17]
- [student5/development-goals/1]

Limitations:
- The analysis is based solely on the provided evidence and does not consider any external factors or additional information.

Reviewer

Status: success

Risk: Low

The worker output accurately matches the evidence records it cites, and the output answers the request and follows the plan. The Automated Checks all PASS, indicating no significant issues. However, the output could be more detailed in explaining the reasoning behind the recommended development activity.

Correction: Add a brief explanation of the analysis and the reasoning behind the recommended development activity.

Retest: Retest the worker output with the requested correction.

Recommendation: The worker output is a good start, but could benefit from a more detailed explanation of the analysis and reasoning behind the recommended development activity.

Human decision: Correct

Reviewed by Matthew Barnard

Correction

The mentoring course supports the goal's requirement to mentor two early-career academics. This remains a proposed activity, not completed training.

```
decision    : Correct
reviewer    : Matthew Barnard                                    Write-Output "Audit events for workflow: $workflowId"
>> rection  : The mentoring course supports the goal's requirement to mentor two early-career academics. This remains a proposed activity,
>> Get-Content -LiteralPath 'ai-services/multi-agent-server/coordination_audit.jsonl' |
>>   ForEach-Object { $_ | ConvertFrom-Json } |
>>   Where-Object { $_.workflow_id -eq $workflowId } |
>>   Select-Object event, agent, status |
>>   Format-Table -AutoSizeeDrive\Documents\AdvSoftware\faculty-management-system-ASD>
Audit events for workflow: 86bd61e8-558b-4a28-a029-c8b3ee5aacc6

event                     agent          status
-----                     -----          ------
workflow_started          coordinator    running
planner_completed         planner_agent  success
evidence_collected        worker_agent   success
worker_completed          worker_agent   success
automated_checks_completed coordinator   passed
reviewer_completed        reviewer_agent success
awaiting_human_review     human_reviewer pending
human_decision_recorded   human_reviewer corrected
```

I submitted a request through the Multi-Agent tab to review staff ID 1's goals and training. The Planner created the plan, my evidence loader supplied recorded development data, and the Worker recommended Mentoring Early-Career Academics. The Reviewer requested clearer reasoning. All three agents and five automated checks succeeded. I selected Correct and explained how the course supported the goal while remaining a proposed activity. The decision appeared in workflow history, while the history and coordination logs recorded the outputs, correction and execution stages under the same workflow ID. The decision did not automatically change the original development records.

11. GitHub Commit Logs

https://github.com/Hein-Htet-Aung08/faculty-management-system-ASD/commits/main/

Andy

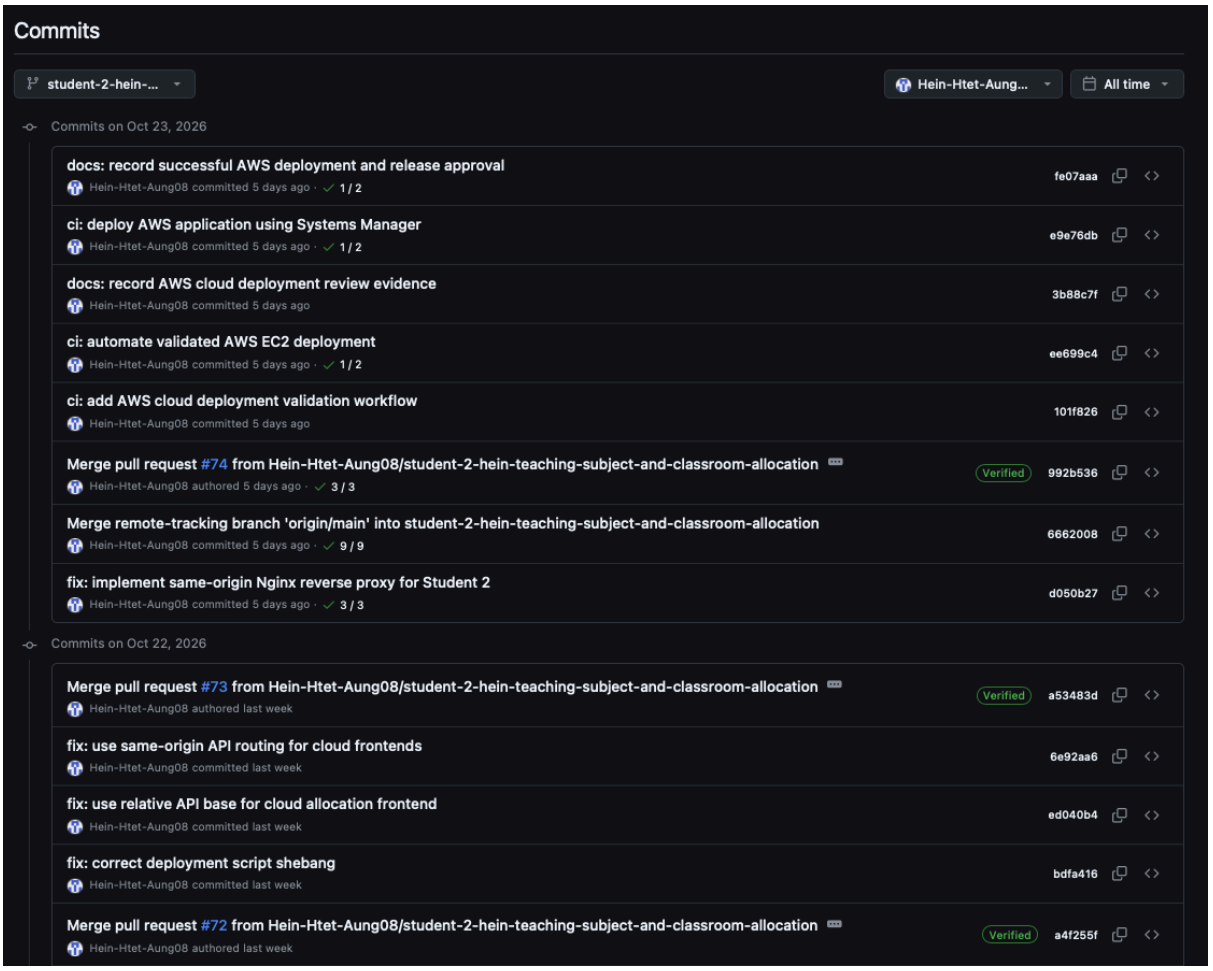
Commits on Oct 22, 2026

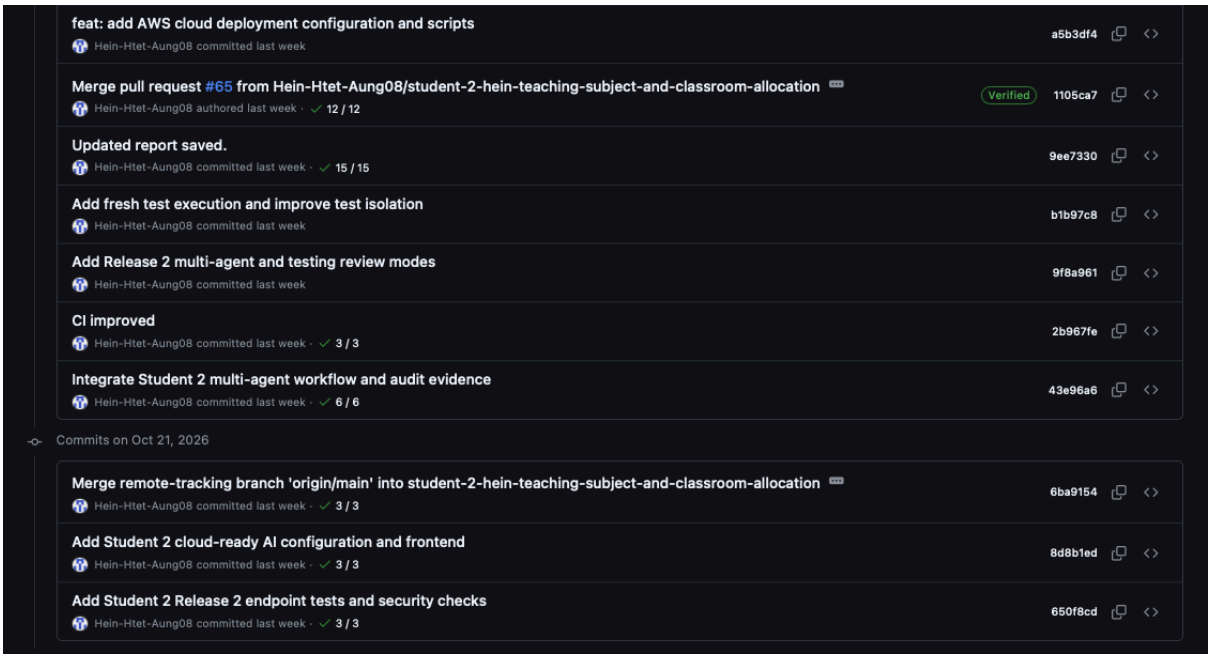
Merge pull request #71 from Hein-Htet-Aung08/student-3-tristan-workload-and-availability-management Alamb-beyonder authored last week · ✓ 16 / 16	Verified	41c1863
Merge branch 'main' into student-3-tristan-workload-and-availability-management Alamb-beyonder authored last week · ✓ 19 / 19	Verified	d47740d
Add Cloud Deployment Report Review mode Alamb-beyonder committed last week		5f9d6a7
Add endpoint tests and pre-commit security scans Alamb-beyonder committed last week · ✓ 13 / 13		2ac62c4
Integrate Staff Management with Multi-Agent Server Alamb-beyonder committed last week · ✓ 4 / 4		bd9cf71
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management Alamb-beyonder committed last week		5618c0e

Commits on Oct 21, 2026

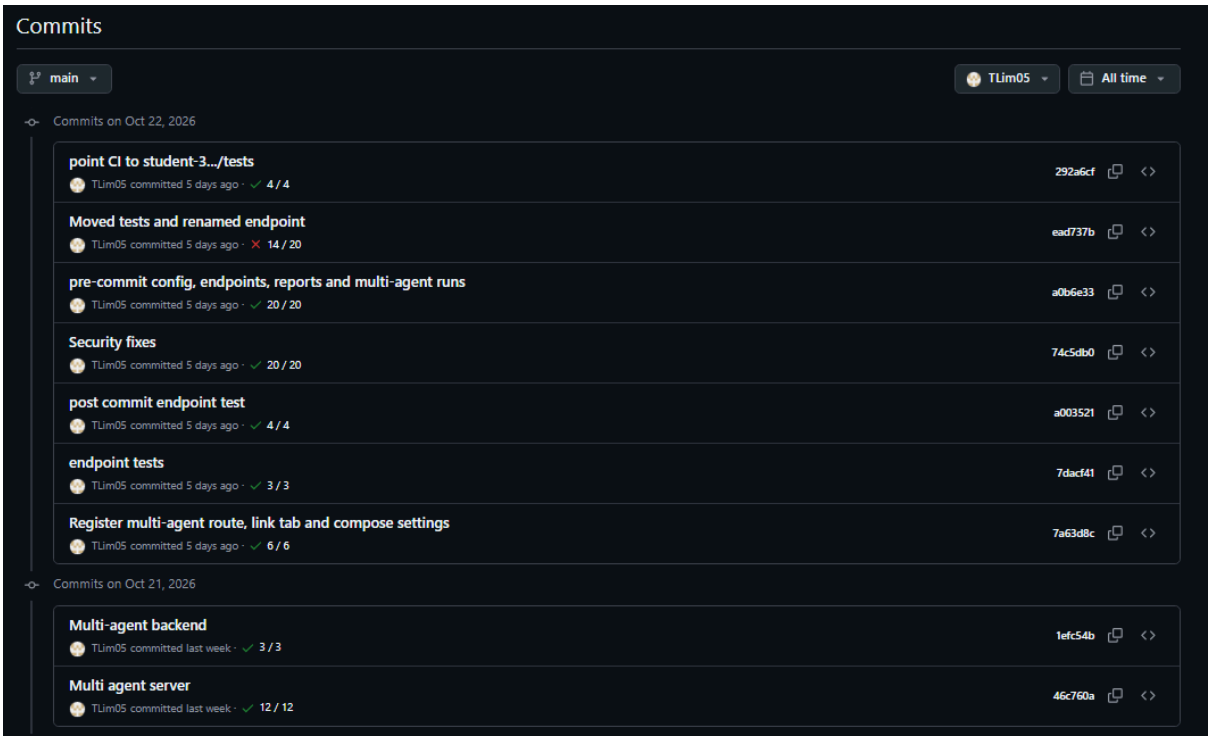
Add AI-Mode toggle and configurable frontend API base Alamb-beyonder committed last week · ✓ 4 / 4		31f1bfb

Hein

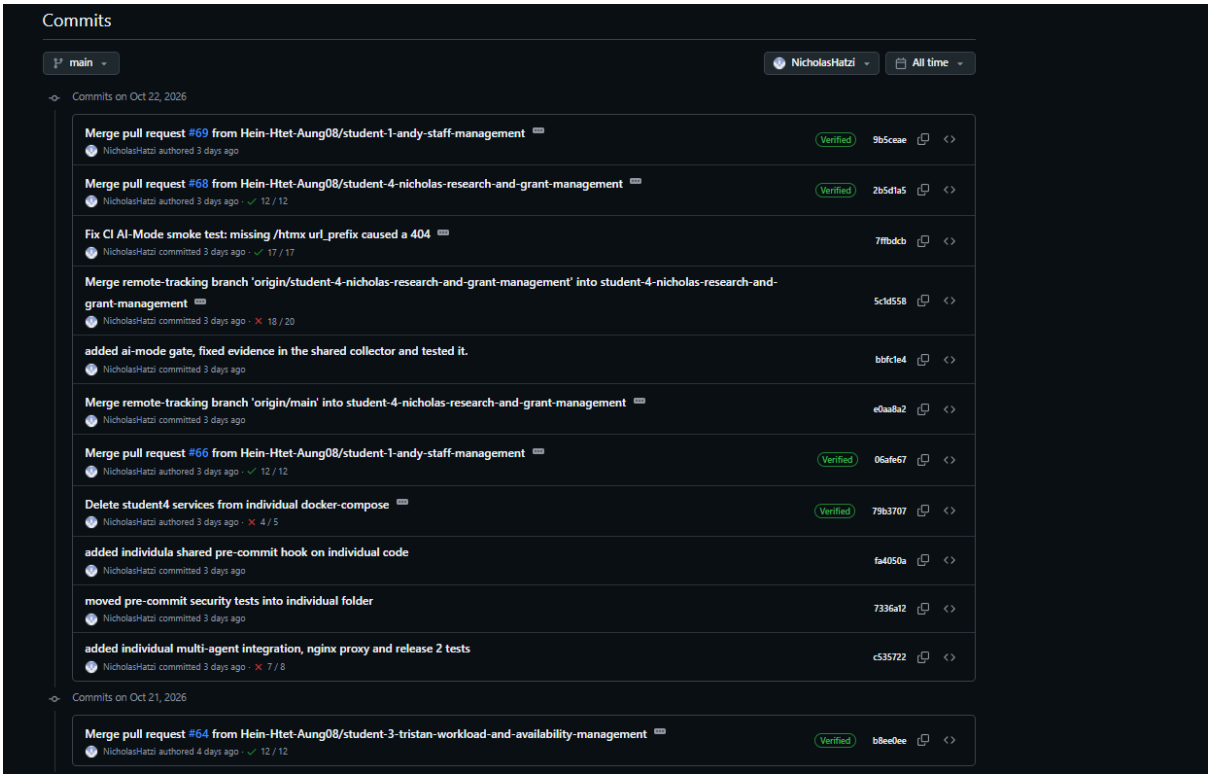

Commits
student-2-hein-...
Hein-Htet-Aung...
All time
Commits on Oct 23, 2026
docs: record successful AWS deployment and release approval
Hein-Htet-Aung08 committed 5 days ago · 1 / 2
fe07aaa
ci: deploy AWS application using Systems Manager
Hein-Htet-Aung08 committed 5 days ago · 1 / 2
e9e76db
docs: record AWS cloud deployment review evidence
Hein-Htet-Aung08 committed 5 days ago
3b88c7f
ci: automate validated AWS EC2 deployment
Hein-Htet-Aung08 committed 5 days ago · 1 / 2
ee699c4
ci: add AWS cloud deployment validation workflow
Hein-Htet-Aung08 committed 5 days ago
101f826
Merge pull request #74 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation
Hein-Htet-Aung08 authored 5 days ago · 3 / 3
Verified
992b536
Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation
Hein-Htet-Aung08 committed 5 days ago · 9 / 9
6662008
fix: implement same-origin Nginx reverse proxy for Student 2
Hein-Htet-Aung08 committed 5 days ago · 3 / 3
d050b27
Commits on Oct 22, 2026
Merge pull request #73 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation
Hein-Htet-Aung08 authored last week
Verified
a53483d
fix: use same-origin API routing for cloud frontends
Hein-Htet-Aung08 committed last week
6e92aa6
fix: use relative API base for cloud allocation frontend
Hein-Htet-Aung08 committed last week
ed040b4
fix: correct deployment script shebang
Hein-Htet-Aung08 committed last week
bdfa416
Merge pull request #72 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation
Hein-Htet-Aung08 authored last week
Verified
a4f255f


feat: add AWS cloud deployment configuration and scripts
Hein-Htet-Aung08 committed last week
a5b3df4
Merge pull request #65 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation
Hein-Htet-Aung08 authored last week · 12 / 12
Verified
1105ca7
Updated report saved.
Hein-Htet-Aung08 committed last week · 15 / 15
9ee7330
Add fresh test execution and improve test isolation
Hein-Htet-Aung08 committed last week
b1b97c8
Add Release 2 multi-agent and testing review modes
Hein-Htet-Aung08 committed last week
9f8a961
CI improved
Hein-Htet-Aung08 committed last week · 3 / 3
2b967fe
Integrate Student 2 multi-agent workflow and audit evidence
Hein-Htet-Aung08 committed last week · 6 / 6
43e96a6
Commits on Oct 21, 2026
Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation
Hein-Htet-Aung08 committed last week · 3 / 3
6ba9154
Add Student 2 cloud-ready AI configuration and frontend
Hein-Htet-Aung08 committed last week · 3 / 3
8d8b1ed
Add Student 2 Release 2 endpoint tests and security checks
Hein-Htet-Aung08 committed last week · 3 / 3
650f8cd

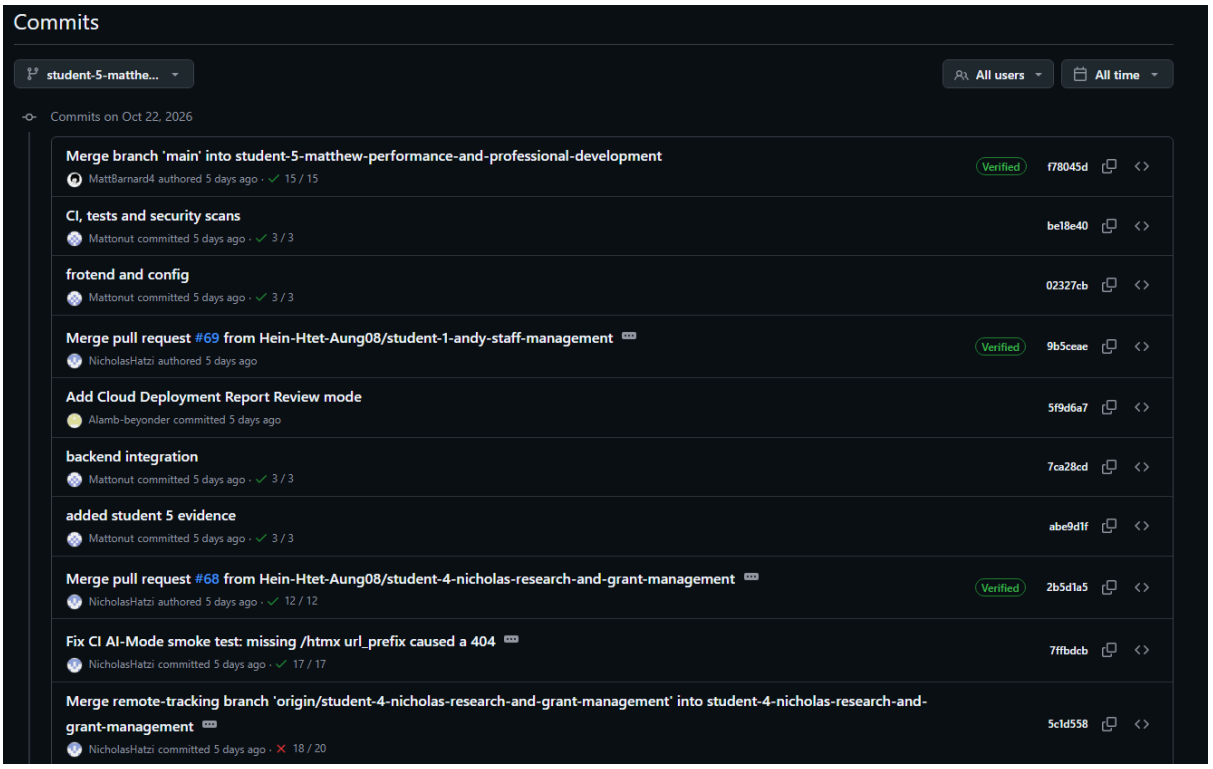
Tristan



Nicholas



Matthew



12. Contribution Logs

Student Number	Name	Contribution
25249309	Andy Lam	• Individual Release 2 Report Sections • Agentic Loop (Cloud Report Review) • Cloud Deployment Review and Report Section • Release 2 Feature Implementation • CI tests for two endpoints and pre-commit security scans
25026511	Hein Htet Aung	• All individual Release 2 report sections and the group Cloud Deployment and CI/CD section • Implementing pre-commit security checks, automated endpoint tests and CI/CD enhancements for individual features • Individual Multi-Agent workflow integration • Agentic Loop enhancements for Multi-Agent Workflow Review and Testing Report Review • Configured AWS EC2 deployment, Docker Compose, Nginx, IAM, SSM and GitHub Actions secrets

		• Automated cloud deployment and performed endpoint validation, deployment review and release verification
25299672	Tristan Lim	• Shared Multi-Agent Server • Individual Multi-Agent workflow • Release 2 Design Flow Diagram • All individual sections of the report.
25471398	Nicholas Hatzidimitriou	• Project Overview section of the report • Testing Review section of the report • Individual Multi-Agent workflow (both Frontend and Backend) • All individual sections of the report and features
25563457	Matthew Barnard	• Local Deployment Evidence section of the report • Individual Muti-Agent workflow (both frontend and backend) • All individual sections of the report and supporting screenshots

13. Showcase Video URL

https://www.youtube.com/watch?v=2VQmOAHN6Ho

14. Github Repository URL

https://github.com/Hein-Htet-Aung08/faculty-management-system-ASD